In ITIL v3/2011, which stage of the Service Lifecycle defines the perspective, position, plans, and patterns that a service provider needs to execute to meet an organization's business outcomes?
A. Service Strategy
B. Service Design
C. Service Transition
D. Service Operation
ANSWER: A

Within ITIL Service Strategy, what are the Four Ps of strategy originally defined by Henry Mintzberg?
A. Planning, Provisioning, Protection, and Pricing
B. People, Processes, Products, and Partners
C. Policy, Portfolio, Pipeline, and Performance
D. Perspective, Position, Plan, and Pattern
ANSWER: D

Which ITIL service provider type is an internal service provider embedded within a specific business unit that delivers services exclusively to that single business unit?
A. Type III Service Provider
B. Type IV Service Provider
C. Type II Service Provider
D. Type I Service Provider
ANSWER: D

How does an ITIL Type II (Shared Services Unit) service provider differ from a Type III service provider?
A. Type II serves only one internal department, whereas Type III serves multiple internal departments
B. Type II provides hardware manufacturing only, whereas Type III provides cloud software only
C. Type II operates without service level agreements, whereas Type III requires statutory compliance audits
D. Type II provides shared IT services to multiple internal business units, whereas Type III is an external provider serving external customers in a competitive market
ANSWER: D

In ITIL Service Strategy, what two synergistic components combine to create the 'Value' of an IT service for a customer?
A. Utility (fit for purpose) and Warranty (fit for use)
B. Cost and Risk
C. Resources and Capabilities
D. Assets and Configurations
ANSWER: A

What are the three main subdivisions that make up the complete Service Portfolio in ITIL Service Portfolio Management?
A. Service Strategy, Service Design, and Service Operation
B. Service Pipeline, Service Catalog, and Retired Services
C. Business Service Catalog, Technical Service Catalog, and Configuration Management Database
D. Pattern of Business Activity, User Profiles, and Service Level Packages
ANSWER: B

Which three core activities are primary responsibilities of Financial Management for IT Services in ITIL Service Strategy?
A. Procurement, Asset Disposal, and Vendor Licensing
B. Capacity Sizing, Demand Forecasting, and Incident Costing
C. Auditing, Tax Filing, and Payroll Processing
D. Budgeting, Accounting, and Charging
ANSWER: D

During peak payroll processing at the end of every month, an organization's ERP infrastructure experiences severe resource contention. Which ITIL Service Strategy process analyzes Patterns of Business Activity (PBA) and User Profiles (UP) to anticipate and regulate this consumption cycle?
A. Availability Management
B. Service Portfolio Management
C. Business Relationship Management
D. Demand Management
ANSWER: D

When a CIO evaluates IT capabilities (management, organization, processes, and knowledge) against

IT resources (financial capital, infrastructure, and applications) to establish a distinct competitive advantage in a market space, which statement accurately reflects ITIL Service Strategy principles?
A. Market spaces are defined solely by physical data center footprints rather than customer outcomes
B. Capabilities represent an organization's ability to coordinate and deploy resources to produce value and are much harder for competitors to duplicate than raw resources
C. Capabilities are tangible balance-sheet hardware assets whereas resources are intangible skills
D. Resources alone guarantee competitive advantage because capabilities are easily purchased off the shelf
ANSWER: B

A newly proposed analytics service is currently under development and has not yet been released to live customers. Which role is accountable for governing the overall Service Portfolio, and in which section of the Service Portfolio must this unleased service be tracked?
A. Demand Manager; tracked in the Definitive Media Library
B. Service Catalog Manager; tracked in the Technical Service Catalog
C. Business Relationship Manager; tracked in Retired Services
D. Service Portfolio Manager; tracked in the Service Pipeline
ANSWER: D

What are the 'Four Ps of Service Design' that must be balanced to ensure effective design and deployment of IT services?
A. Policies, Procedures, Protocols, and Packages
B. Price, Place, Promotion, and Product
C. Perspective, Position, Plan, and Pattern
D. People, Processes, Products, and Partners
ANSWER: D

Which document produced during the ITIL Service Design stage defines all aspects of an IT service and its requirements through every subsequent stage of its lifecycle?
A. Request for Change (RFC)
B. Service Design Package (SDP)
C. Service Level Agreement (SLA)
D. Service Improvement Plan (SIP)
ANSWER: B

In ITIL Information Security Management (ISM), what does the CIA triad stand for?
A. Confidentiality, Integrity, and Availability
B. Configuration, Integration, and Automation
C. Controls, Incidents, and Audits
D. Compliance, Identity, and Authentication
ANSWER: A

How does the Business/Customer Service Catalog differ from the Technical/Supporting Service Catalog in ITIL Service Catalog Management?
A. The Business Service Catalog is accessible only to third-party vendors, whereas the Technical Service Catalog is public
B. The Business Service Catalog tracks financial budgets, whereas the Technical Service Catalog tracks user passwords
C. The Business Service Catalog contains details of live IT services delivered to customers, whereas the Technical Service Catalog details supporting services, shared components, and CIs underpinning those customer-facing services
D. The Business Service Catalog lists retired services, whereas the Technical Service Catalog lists pipeline services
ANSWER: C

An internal IT Network Team signs an agreement with the internal Database Administration Team committing to restore failed core switch links within 30 minutes so the company can meet its customer-facing SLA. What type of agreement is this?
A. Underpinning Contract (UC)
B. Service Level Requirement (SLR)
C. Operational Level Agreement (OLA)
D. Service Charter
ANSWER: C

What are the three interconnected sub-processes of Capacity Management in ITIL Service Design?
A. Business Capacity Management, Service Capacity Management, and Component Capacity Management
B. Network Capacity, Storage Capacity, and Compute Capacity Management

C. Strategic Capacity, Tactical Capacity, and Operational Capacity Management
D. Event Capacity, Incident Capacity, and Problem Capacity Management
ANSWER: A

In ITIL IT Service Continuity Management (ITSCM), which metric defines the maximum tolerable duration that an IT service can remain unavailable after a major disruption before severe business damage occurs?
A. Mean Time To Repair (MTTR)
B. Recovery Point Objective (RPO)
C. Recovery Time Objective (RTO)
D. Mean Time Between System Incidents (MTBSI)
ANSWER: C

During a Business Impact Analysis (BIA) for a banking application, stakeholders specify that no more than 15 minutes of committed database transactions can be lost in the event of a catastrophic storage failure. Which continuity metric does this 15-minute window represent, and what mechanism directly satisfies it?
A. Service Level Objective (SLO); satisfied by an Operational Level Agreement
B. Recovery Time Objective (RTO); satisfied by warm standby server provisioning
C. Mean Time Between Failures (MTBF); satisfied by dual power supplies
D. Recovery Point Objective (RPO); satisfied by synchronous or frequent near-real-time database replication and log shipping
ANSWER: D

Which five major aspects of Service Design must be comprehensively addressed when designing a new or modified IT service?
A. Financials, Portfolios, Demand, Suppliers, and Security
B. Incidents, Problems, Changes, Releases, and Configurations
C. Strategy, Design, Transition, Operation, and Continual Improvement
D. Service Solutions, Management Information Systems and Tools, Technology and Management Architectures, Processes, and Measurement Methods and Metrics
ANSWER: D

A global enterprise delivers four distinct IT services to three separate business units. If every business unit requires identical service targets across all four IT services, which SLA structure is most efficient to minimize administrative overhead without duplicating documents?
A. Service-based SLA
B. Customer-based SLA
C. Ad-hoc Underpinning SLA
D. Multi-level SLA
ANSWER: D

In ITIL Service Asset and Configuration Management (SACM), what term refers to any hardware, software, documentation, or service component that needs to be managed in order to deliver an IT service?
A. Configuration Item (CI)
B. Service Request (SR)
C. Known Error (KE)
D. Standard Change (SC)
ANSWER: A

What is the hierarchical four-tier structure used in ITIL Knowledge Management to transform raw facts into actionable decisions?
A. People, Process, Products, Partners (4Ps)
B. Events, Incidents, Problems, Changes (EIPC)
C. Data-to-Information-to-Knowledge-to-Wisdom (DIKW)
D. Plan, Do, Check, Act (PDCA)
ANSWER: C

Which organizational entity in ITIL Service Operation serves as the Single Point of Contact (SPOC) between the service provider and users for day-to-day operational communications?
A. Change Advisory Board (CAB)
B. Technical Management
C. Service Desk
D. Transition Planning and Support
ANSWER: C

How does ITIL Service Operation define an 'Incident'?
A. A formal proposal to modify a baseline Configuration Item in production

B. An unplanned interruption to an IT service or a reduction in the quality of an IT service
C. The underlying unknown root cause of one or more recurring infrastructure failures
D. Any state change that has significance for the management of a Configuration Item
ANSWER: B

Which category of change in ITIL Change Management is pre-authorized, low-risk, relatively common, and follows an established documented procedure or work instruction without requiring CAB review each time?
A. Retrospective Change
B. Emergency Change
C. Standard Change
D. Normal Change
ANSWER: C

How do the Definitive Media Library (DML) and the Configuration Management Database (CMDB) interact during ITIL Service Transition?
A. The DML stores physical hardware spare parts, whereas the CMDB stores user credentials
B. The DML securely stores master copies of authorized software media and licenses, while the CMDB stores logical records and attributes representing those media items as Configuration Items
C. The DML replaces the CMDB in cloud-native environments
D. The CMDB stores raw source code binaries, whereas the DML stores incident tickets
ANSWER: B

During Release and Deployment Management, a financial firm rolls out a new trading platform to a single regional branch office first to validate stability in a live environment before deploying it company-wide. What deployment approach is being used?
A. Pilot deployment
B. Shadow parallel retirement
C. Big Bang deployment
D. Push-only automated deployment
ANSWER: A

In ITIL Event Management, what are the three primary classifications of events generated by CIs and monitoring tools?
A. Informational, Warning, and Exception events
B. Standard, Normal, and Emergency events
C. Hardware, Software, and Network events
D. Low, Medium, and High priority events
ANSWER: A

In ITIL Problem Management, where are documented root causes (Known Errors) and their corresponding temporary workarounds stored so the Service Desk can resolve recurring incidents faster?
A. Supplier and Contract Management Information System (SCMIS)
B. Definitive Media Library (DML)
C. Known Error Database (KEDB)
D. Service Portfolio Pipeline
ANSWER: C

A user submits a request through a self-service portal to receive a standard read-only access token to a reporting folder and have a pre-approved desktop utility installed. Which ITIL Service Operation process handles this routine fulfillment?
A. Emergency Change Management
B. Incident Management
C. Problem Management
D. Request Fulfillment
ANSWER: D

In ITIL Service Operation, operational teams continually face four fundamental sets of conflicting goals (such as Internal IT view vs. External business view, and Stability vs. Responsiveness). How should a Service Operation Manager resolve the conflict between 'Quality of Service' and 'Cost of Service'?
A. Always prioritize cutting operational costs by eliminating redundancy regardless of SLA penalties
B. Always over-provision hardware to 99.999% availability regardless of budget constraints
C. Align operational delivery strictly to the agreed Service Level Agreements (SLAs) and business value, ensuring neither excessive under-delivery nor unauthorized gold-plating occurs
D. Freeze all software changes permanently to maximize stability at zero deployment cost
ANSWER: C

A critical zero-day vulnerability is actively exploiting a production payment gateway at 02:00 AM, requiring an immediate firewall rule and kernel patch change. Which body has the authority to evaluate and authorize this high-impact modification rapidly?
A. The Knowledge Management Process Owner
B. The Emergency Change Advisory Board (ECAB)
C. The Service Desk Analyst on shift
D. The Continual Service Improvement Register Manager
ANSWER: B

In ITIL Service Operation, what is the primary distinction between the Technical Management function and the Application Management function?
A. Technical Management operates the Level 1 Service Desk, whereas Application Management operates physical data center security
B. Technical Management provides custodian expertise and operational support for the IT infrastructure (servers, OS, networks, storage), whereas Application Management manages software applications throughout their lifecycle
C. Technical Management only works during Service Design, whereas Application Management only works during Service Transition
D. Technical Management handles third-party vendor contracts, whereas Application Management handles financial accounting
ANSWER: B

Multiple large release packages from three separate development teams are scheduled for deployment during the same maintenance window, creating contention for test environments and shared database administrators. Which ITIL Service Transition process and role coordinate resources and resolve scheduling conflicts across these multiple transitions?
A. Problem Management led by the Problem Manager
B. Incident Management led by the Major Incident Manager
C. Transition Planning and Support led by the Transition Planning and Support Process Manager
D. Service Asset and Configuration Management led by the Configuration Librarian
ANSWER: C

Which four-stage quality improvement cycle, popularized by W. Edwards Deming, forms the foundational methodology for ITIL Continual Service Improvement (CSI)?
A. Identify, Contain, Eradicate, Recover
B. Strategy, Design, Transition, Operate
C. Define, Measure, Analyze, Control
D. Plan, Do, Check, Act (PDCA)
ANSWER: D

Where are all identified improvement opportunities, along with their priority, categorization, and expected business benefits, centrally recorded and tracked in ITIL CSI?
A. Definitive Media Library (DML)
B. Continual Service Improvement (CSI) Register
C. Configuration Management Database (CMDB)
D. Known Error Database (KEDB)
ANSWER: B

Which role is ultimately accountable for the success of all Continual Service Improvement activities and for maintaining the CSI Register across the organization?
A. Network Operations Technician
B. Service Desk Supervisor
C. CSI Manager
D. Release Packaging Administrator
ANSWER: C

In the ITIL CSI Model (the six-question framework), which question immediately follows 'Where are we now?' (Baseline assessment)?
A. Did we get there?
B. What is the vision?
C. How do we get there?
D. Where do we want to be?
ANSWER: D

What are the three main types of metrics that ITIL Continual Service Improvement collects and analyzes to evaluate performance?
A. Technology metrics, Process metrics, and Service metrics
B. Local metrics, Regional metrics, and Global metrics

C. Hardware metrics, Software metrics, and Facilities metrics
D. Financial metrics, HR metrics, and Marketing metrics
ANSWER: A

In the RACI authority matrix used in ITIL CSI to assign clear process ownership and responsibilities, what do the letters R, A, C, and I stand for?
A. Reviewed, Audited, Certified, and Inspected
B. Responsible, Accountable, Consulted, and Informed
C. Reliable, Available, Confidential, and Integrated
D. Requested, Approved, Configured, and Implemented
ANSWER: B

Why are structured training and organizational awareness campaigns critical during Continual Service Improvement initiatives?
A. To eliminate the need for Service Level Agreements with business customers
B. To overcome organizational resistance to change, build staff competencies in updated processes, and embed a culture of ongoing quality improvement
C. To transfer process accountability from the Process Owner to external end users
D. To replace automated monitoring tools with manual spreadsheet logging
ANSWER: B

When assigning roles using a RACI chart for an ITIL CSI improvement initiative, which rule must strictly be enforced to prevent governance failure?
A. No activity is allowed to have both a Responsible (R) and an Accountable (A) role assigned simultaneously
B. Consulted (C) communication is strictly one-way, whereas Informed (I) requires two-way consensus
C. Exactly one role must be assigned as Accountable (A) for each activity, although multiple roles may be Responsible (R)
D. Every activity must have at least three different people assigned as Accountable (A)
ANSWER: C

In the ITIL 7-Step Improvement Process, what is the exact sequence of the first four steps?
A. 1. Implement improvement -> 2. Analyze the information -> 3. Gather the data -> 4. Define what you could measure
B. 1. Present and use the information -> 2. Define what you will measure -> 3. Identify the strategy -> 4. Gather the data
C. 1. Gather the data -> 2. Process the data -> 3. Analyze the information -> 4. Present the information
D. 1. Identify the strategy for improvement -> 2. Define what you will measure -> 3. Gather the data -> 4. Process the data
ANSWER: D

An IT organization wants to establish an ongoing schedule of service reviews and maturity assessments under CSI, but they failed to record initial performance metrics before implementing a new incident automation tool. According to the CSI model, what is the primary consequence of failing to establish a baseline?
A. The CMDB will automatically reject all subsequent Request for Change tickets
B. The Service Desk must shut down until a third-party ISO audit is completed
C. All Technology metrics are automatically converted into Service metrics
D. There is no objective reference point ('Where are we now?') to quantify whether the improvement initiative actually achieved its target ('Did we get there?')
ANSWER: D

According to the Uptime Institute Data Center Tier Standard, which Tier level provides a 'Fault Tolerant' site infrastructure with multiple active power and cooling distribution paths and an expected availability of 99.995%?
A. Tier III
B. Tier I
C. Tier IV
D. Tier II
ANSWER: C

In standard data center server racks, how tall is one Rack Unit (1U)?
A. 3.50 inches (88.9 mm)
B. 1.00 inch (25.4 mm)
C. 2.50 inches (63.5 mm)
D. 1.75 inches (44.45 mm)
ANSWER: D

What is the primary role of a Network Operations Center (NOC) in data center management?
A. Conducting annual financial tax audits for corporate shareholders
B. Manufacturing silicon microprocessors for server motherboards
C. Writing application source code for customer mobile apps
D. Continuous 24x7 monitoring, event correlation, incident triage, and performance management of network, server, and facility infrastructure
ANSWER: D

Which industry metric is used to measure the energy efficiency of a data center by dividing the Total Facility Power by the IT Equipment Power?
A. Mean Time Between Failures (MTBF)
B. British Thermal Units per Hour (BTU/hr)
C. Uninterruptible Power Factor (UPF)
D. Power Usage Effectiveness (PUE)
ANSWER: D

Why do modern data centers implement a Hot Aisle / Cold Aisle rack layout with physical aisle containment?
A. To convert AC power directly into DC power without using PDUs
B. To reduce the length of fiber optic patch cables between switches
C. To prevent the mixing of chilled supply air and hot exhaust air, thereby improving CRAC/CRAH cooling efficiency and preventing server thermal throttling
D. To allow unauthenticated visitors to walk directly behind server racks
ANSWER: C

In structured modular data center cabling (ANSI/TIA-942), which area contains the core routers and switches that distribute backbone cabling out to the Horizontal Distribution Areas (HDAs) and Zone Distribution Areas (ZDAs)?
A. Equipment Distribution Area (EDA)
B. Telecommunications Closet (TC)
C. Main Distribution Area (MDA)
D. Entrance Room (ER)
ANSWER: C

To ensure carrier redundancy and eliminate single points of failure for WAN connectivity, how should a Tier III/IV data center connect to upstream Internet Service Providers (ISPs)?
A. Use a single fiber cable connected to two different routers in the same rack
B. Use Wi-Fi bridges as the primary WAN link for production database replication
C. Subscribe to at least two independent ISPs entering the building through physically diverse Entrance Rooms (points of entry) using BGP multi-homing
D. Route all ISP links through a single shared street manhole outside the facility
ANSWER: C

Which physical security mechanism prevents 'tailgating' (unauthorized individuals following an authorized badge holder into a high-security data center hall) by requiring one-person-at-a-time verification between two interlocking doors?
A. Biometric Mantrap (Access Control Vestibule)
B. Bollard perimeter fencing
C. VESDA aspirating smoke detector
D. Closed-Circuit Television (CCTV) camera
ANSWER: A

When selecting a geographic location for a primary enterprise data center, which combination of environmental and infrastructure factors represents the best planning practice?
A. Building exclusively on the top floor of a downtown retail mall with shared passenger elevators
B. Selecting a low-risk natural disaster zone (low seismic/flood risk) with access to dual independent power grids, diverse carrier fiber backbones, and favorable ambient climate
C. Choosing a remote location with a single rural power utility and satellite-only internet
D. Locating in a seismic fault zone next to a chemical plant to lower real estate taxes
ANSWER: B

A data center draws 1,500 kW of total utility power across the entire facility (including chillers, lighting, UPS losses, and IT racks), while the servers, storage arrays, and network switches consume 1,000 kW. What is the facility's Power Usage Effectiveness (PUE), and what does an ideal theoretical PUE value approach?
A. PUE is 500 kW; ideal theoretical value approaches 100 kW
B. PUE is 0.67; ideal theoretical value approaches 0.0
C. PUE is 2.50; ideal theoretical value approaches 2.0

D. PUE is 1.50; ideal theoretical value approaches 1.0
ANSWER: D

Why is specialized sub-floor and rack cleaning (using HEPA-filtered vacuums rather than dry sweeping or water mops) alongside clean-agent fire suppression (such as FM-200 or Novec 1230) mandatory in a production data center?
A. HEPA vacuums increase the humidity above 95% to prevent electrostatic discharge
B. FM-200 floods the server room with foam to cool down overheating CPUs during peak load
C. Particulate dust and zinc whiskers under raised floors can clog server heatsinks or cause electrical short circuits, while clean-agent gas extinguishes fires without conductive water damage to live electronics
D. Sub-floor dust interferes with BGP routing tables on logical VLAN interfaces
ANSWER: C

When capacity planners calculate the required physical area and network bandwidth for a new data center hall, why must they allocate dedicated unoccupied space ('white space' expansion headroom and staging areas) and redundant 2N or N+1 capacity?
A. To ensure the data center operates at a PUE greater than 3.5 at all times
B. To permit future rack growth, provide safe aisle clearance for equipment replacement and airflow, and support concurrent maintenance without interrupting live workloads
C. To satisfy municipal parking regulations for office employees
D. Because standard 42U racks cannot hold more than 5 servers due to Ethernet cable weight
ANSWER: B

In virtualization technology, what software or firmware layer abstracts physical host hardware (CPU, memory, storage, and network) and enables multiple isolated Virtual Machines (VMs) to run concurrently on a single physical machine?
A. Device Driver Manager
B. Hypervisor (Virtual Machine Monitor)
C. Logical Volume Manager
D. Container Registry
ANSWER: B

Which of the following is an example of a Type-1 (bare-metal) hypervisor that installs directly on physical server hardware without requiring an underlying host operating system?
A. Parallels Desktop for Mac
B. Oracle VirtualBox
C. VMware Workstation Player
D. VMware ESXi
ANSWER: D

How does a Type-2 (hosted) hypervisor operate compared to a Type-1 hypervisor?
A. A Type-2 hypervisor can only virtualize network switches and not CPU or RAM
B. A Type-2 hypervisor requires dedicated mainframe hardware to boot guest VMs
C. A Type-2 hypervisor runs directly on the server's BIOS/UEFI firmware without any operating system
D. A Type-2 hypervisor runs as an application layer on top of a conventional host operating system such as Windows, macOS, or Linux
ANSWER: D

What is the fundamental architectural difference between Full Hardware-Assisted Virtualization and Para-Virtualization?
A. Full virtualization requires modifying the guest OS source code, whereas para-virtualization runs unmodified guest operating systems
B. Para-virtualization emulates every CPU instruction purely in software without hypervisor involvement
C. Full virtualization shares the host OS kernel across containers, whereas para-virtualization uses separate physical servers
D. Full hardware-assisted virtualization uses CPU extensions (such as Intel VT-x or AMD-V) to run an unmodified guest OS in complete isolation, whereas para-virtualization modifies the guest OS kernel to make direct 'hypercalls' to the hypervisor
ANSWER: D

What exactly is captured when an administrator takes a 'Snapshot' of a Virtual Machine?
A. Only the network MAC address and firewall rules of the physical hypervisor
B. The point-in-time state, disk data delta, configuration, and optionally the active memory state of the VM at the exact moment the snapshot was taken
C. A physical RAID mirror of the entire SAN storage array
D. A full bootable golden master image pushed to a remote cloud object store

ANSWER: B

In virtualization management, what is the difference between a 'Full Clone' and a 'Linked Clone' of a Virtual Machine?
A. A Full Clone only copies RAM state, whereas a Linked Clone copies the OS kernel
B. A Full Clone shares its virtual disk continuously with the parent VM, whereas a Linked Clone is an independent copy
C. A Linked Clone can migrate across separate storage arrays without access to the parent disk, whereas a Full Clone cannot
D. A Full Clone is a completely independent copy of the VM that shares nothing with the parent after creation, whereas a Linked Clone shares the parent VM's base virtual disk (via a snapshot) and stores only differential changes
ANSWER: D

Why do system administrators convert a configured, hardened Virtual Machine into a 'VM Template'?
A. To compress the hypervisor kernel so it fits on a USB flash drive
B. To bypass CPU virtualization extensions on legacy motherboards
C. To allow users to log into the template directly as a live production database server
D. To create an immutable, standardized golden master image that cannot be powered on directly and is used to provision new, consistent VMs rapidly
ANSWER: D

Why is it considered a critical anti-pattern in production virtualization to rely on VM snapshots as long-term backups or keep long chains of snapshots active for months?
A. Snapshot delta files grow continuously as blocks change, degrading disk I/O performance, consuming datastore capacity, and remains dependent on the base disk on the same storage volume
B. Snapshots disable Intel VT-x and AMD-V hardware acceleration across the entire cluster
C. A VM with an active snapshot cannot receive network traffic on TCP port 443
D. Snapshots automatically format the guest OS root partition after 30 days
ANSWER: A

When provisioning 50 new Windows or Linux VMs from a single VM Template in an enterprise environment, what customization steps must occur during deployment to prevent network and directory conflicts?
A. The hypervisor must convert all 50 VMs into Type-2 hypervisors before booting
B. All 50 VMs must keep the exact same MAC address, hostname, IP address, and Windows Security Identifier (SID)
C. A guest customization specification (or cloud-init/Sysprep) must generate unique MAC addresses, hostnames, IP configurations, SSH host keys, and machine SIDs/UUIDs for each deployed VM
D. The administrator must recompile the guest kernel from C source code on all 50 VMs manually
ANSWER: C

On an x86-64 architecture using Intel VT-x or AMD-V hardware-assisted virtualization, how does the processor resolve the classic Ring 0 privilege conflict between the Hypervisor and the Guest OS kernel?
A. It forces the Guest OS kernel to run in Ring 3 alongside unprivileged user applications
B. It introduces a new privilege mode (VMX Root Operation / Ring -1 for the hypervisor and VMX Non-Root Operation for the guest), allowing the guest OS kernel to run natively in its own Ring 0 while trapping sensitive instructions to the hypervisor
C. It disables virtual memory paging and runs all VMs in 16-bit Real Mode
D. It translates every x86 instruction into ARM assembly at runtime using binary translation
ANSWER: B

What is the fundamental difference between a Storage Area Network (SAN) and Network Attached Storage (NAS)?
A. SAN only works over wireless networks, whereas NAS requires coaxial cables
B. SAN cannot use RAID or ZFS, whereas NAS only supports magnetic tape drives
C. SAN provides file-level storage using NFS/SMB, whereas NAS provides block-level storage using Fibre Channel/iSCSI
D. SAN provides block-level storage that appears to the client OS as a locally attached disk (via iSCSI or Fibre Channel), whereas NAS provides file-level shared storage over a network (via NFS or SMB/CIFS)
ANSWER: D

FreeNAS (now known as TrueNAS CORE) is built on which underlying open-source operating system and defaults to which enterprise file system and volume manager?
A. Ubuntu Linux with the ext4 file system
B. Red Hat Enterprise Linux with the XFS file system
C. Windows Server Core with the NTFS file system

D. FreeBSD with the OpenZFS (ZFS) file system
ANSWER: D

Which IP-based storage protocol encapsulates SCSI block-storage commands inside TCP/IP packets so servers can access remote SAN storage over standard Ethernet networks?
A. SMB (Server Message Block)
B. iSCSI (Internet Small Computer Systems Interface)
C. FTP (File Transfer Protocol)
D. NFS (Network File System)
ANSWER: B

In an iSCSI Storage Area Network architecture, what are the standard terms for the client server requesting block storage and the storage array providing the block storage, as well as the default TCP port used?
A. The client is the iSCSI Target, the storage server is the iSCSI Initiator, and the default port is TCP 2049
B. The client is the iSCSI Initiator, the storage server is the iSCSI Target, and the default port is TCP 3260
C. The client is the iSCSI Broker, the storage server is the iSCSI Node, and the default port is UDP 161
D. The client is the LUN Master, the storage server is the LUN Slave, and the default port is TCP 445
ANSWER: B

In ZFS volume configuration on FreeNAS/TrueNAS, which RAID-Z level requires a minimum of 4 disks and can survive the simultaneous failure of any 2 disks within the vdev without data loss?
A. RAID-Z1
B. RAID-Z3
C. RAID-Z2
D. Stripe (RAID 0)
ANSWER: C

Which command-line utility is used in ZFS to create, destroy, and check the health status of storage pools (such as zpool status), versus the utility used to manage ZFS datasets, quotas, compression, and snapshots?
A. mdadm for pools and mount for datasets
B. zpool for storage pools and zfs for datasets/volumes
C. fdisk for pools and mkfs for datasets
D. pvcreate for pools and lvcreate for datasets
ANSWER: B

Why is shared SAN block storage essential for enabling High Availability (HA) and live virtual machine migration (such as VMware vMotion or KVM live migration) across a hypervisor cluster?
A. Because multiple hypervisor hosts can concurrently access the same shared LUNs holding VM disk files, allowing another host to immediately take over or migrate a running VM if one physical host fails
B. Because SAN storage replaces the need for RAM inside the physical hypervisor hosts
C. Because SAN arrays automatically recompile application binaries into WebAssembly
D. Because SAN eliminates the need for TCP/IP networking between web browsers and servers
ANSWER: A

In a high-availability SAN environment, a server connects to an iSCSI storage array through two Dual-Port NICs across two separate switches. Which Linux technology aggregates these multiple physical network paths to provide both active/passive or active/active path failover and load balancing to the same LUN?
A. Dynamic Host Configuration Protocol (DHCP)
B. Spanning Tree Protocol (STP)
C. Network Address Translation (NAT)
D. Device Mapper Multipathing (DM-Multipath / Multipath I/O - MPIO)
ANSWER: D

How does ZFS protect against 'silent data corruption' (bit rot) and the 'RAID-5 write hole' problem?
A. By using a Copy-on-Write (CoW) transactional object model combined with end-to-end 256-bit checksums stored in parent block pointers (Merkle tree) that automatically self-heal corrupted blocks using redundant vdev parity/mirrors during reads or zpool scrub
B. By requiring every client machine to maintain a local tape backup of the entire pool
C. By running defragmentation tasks every hour and disabling write caching
D. By formatting all drives as FAT32 before adding them to the pool

ANSWER: A

When configuring an iSCSI block share in FreeNAS/TrueNAS for an enterprise hypervisor, which ZFS object is created to expose a fixed-size block device as an iSCSI Extent/LUN, and what authentication mechanism is commonly configured on the Portal/Target to restrict unauthorized initiators?
A. A ZFS Volume (Zvol) exposed as a Device Extent, secured via Initiator IQN ACLs and Challenge-Handshake Authentication Protocol (CHAP / Mutual CHAP)
B. An NFSv3 export secured via /etc/hosts.allow only
C. An SMB workgroup share secured via NetBIOS broadcast
D. A ZFS Directory Share exposed with anonymous FTP login
ANSWER: A

According to the NIST definition of Cloud Computing, which of the following lists the five essential characteristics of cloud computing?
A. Local storage, Tape backups, Dedicated copper lines, Manual patching, and Static IP routing
B. On-demand self-service, Broad network access, Resource pooling, Rapid elasticity, and Measured service
C. Virtualization, Containerization, Orchestration, Compilation, and Debugging
D. High cost, Manual provisioning, Single-tenant hardware, Fixed capacity, and Annual billing
ANSWER: B

In Cloud Computing terminology, what does the 'SPI' in the Cloud SPI Model stand for?
A. Storage, Processing, and Networking
B. Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS)
C. Security, Privacy, and Identity
D. Scalability, Performance, and Integration
ANSWER: B

Which cloud deployment model combines an organization's on-premises Private Cloud infrastructure with a third-party Public Cloud, enabling data and application portability between them?
A. Isolated Air-Gapped Cloud
B. Single-Tenant Local Cloud
C. Community Cloud
D. Hybrid Cloud
ANSWER: D

In the Cloud Shared Responsibility Model, how does customer responsibility compare across IaaS, PaaS, and SaaS?
A. The customer manages the physical data center cabling in SaaS, while the provider manages application data in IaaS
B. The cloud provider is 100% responsible for user password strength and data classification in all three models
C. There is no difference in customer operational responsibility between IaaS, PaaS, and SaaS
D. In IaaS, the customer manages the OS, middleware, runtime, applications, and data; in PaaS, the customer primarily manages their application code and data; in SaaS, the customer primarily manages user access and their own data/configuration
ANSWER: D

In Cloud Security, what is the primary role of Identity and Access Management (IAM), and which security principle should govern all IAM permission assignments?
A. IAM encrypts submarine fiber cables; governed by the Principle of Open Access
B. IAM controls authentication (who can sign in) and authorization (what resources they can access); governed by the Principle of Least Privilege
C. IAM negotiates vendor pricing discounts; governed by the Principle of Shared Billing
D. IAM monitors CPU temperature; governed by the Principle of Maximum Throughput
ANSWER: B

A cloud provider's Service Level Agreement (SLA) guarantees 99.9% monthly uptime for a compute service. If the provider breaches this uptime commitment during a billing cycle, what is the standard remedy specified in a cloud SLA?
A. The cloud provider transfers legal ownership of its physical data center to the customer
B. The customer receives financial service credits applied toward future cloud billing statements according to the SLA penalty tier
C. The cloud provider rewrites the customer's legacy source code for free
D. The customer is exempted from all national data privacy regulations
ANSWER: B

When mapping core enterprise workloads to cloud service categories (Compute, Database, Storage, Developer Tools, Security, and Integration), which combination accurately matches each cloud service type to its primary function?
A. Compute for running VMs/containers/serverless functions; Managed Relational/NoSQL Databases for structured/semi-structured queries; Object/Block Storage for unstructured blobs and attached volumes; Integration (Message Queues/Event Buses) for decoupling microservices
B. Mobile services for BIOS firmware flashing; Web services for crimping RJ45 cables
C. Object Storage for running SQL joins; Compute VMs for storing static cold archives; IAM for video transcoding
D. Developer Tools for physical rack cooling; Media services for DNS resolution
ANSWER: A

In cloud architecture, what is the difference between Horizontal Scaling (scaling out/in) and Vertical Scaling (scaling up/down)?
A. Horizontal scaling requires taking the entire application offline for several hours, whereas vertical scaling is always stateless
B. Horizontal scaling adds or removes instances/nodes in a distributed pool behind a load balancer, whereas vertical scaling increases or decreases the compute resources (CPU/RAM) of an individual instance
C. Horizontal scaling adds more CPU/RAM to a single existing server, whereas vertical scaling adds more server instances to a pool
D. Horizontal scaling only applies to magnetic hard drives, whereas vertical scaling only applies to network cables
ANSWER: B

According to cloud-native development best practices (such as the Twelve-Factor App methodology), how should web application instances manage user session state and configuration settings to ensure seamless auto-scaling?
A. Store all application logs exclusively in volatile RAM without shipping them to a centralized logging service
B. Store user sessions in local /tmp files on the instance boot disk and hardcode database passwords inside compiled source files
C. Design compute instances to be stateless by externalizing session data to a distributed in-memory cache (such as Redis/Memcached) or database, and inject environment-specific configuration via environment variables or secret managers
D. Bind each user permanently to a specific server MAC address and disable load balancing
ANSWER: C

When a backend microservice integrates with a Cloud Provider's RESTful API to dynamically provision storage or invoke a service, what is the secure, production-grade mechanism for authenticating API requests, and how should transient HTTP 429 (Too Many Requests) or 503 responses be handled?
A. Send API requests via SMTP email attachments and wait 24 hours for manual operator approval
B. Embed the root account's master password in plain text inside client-side JavaScript and retry failed requests in an infinite tight loop without delay
C. Use anonymous unencrypted HTTP requests over port 80 and reboot the server whenever a 429 status code occurs
D. Use short-lived, role-based IAM credentials/tokens (such as OAuth2 bearer tokens or HMAC-signed request headers via the cloud SDK) and implement idempotent retries with exponential backoff and jitter
ANSWER: D

An e-commerce company deploys its cloud application across multiple Availability Zones (AZs) within a single geographic Region using an Application Load Balancer and a Multi-AZ relational database. What failure scenario does this architecture protect against, and what scenario would still require a Multi-Region recovery architecture?
A. It protects against the failure of an entire physical data center facility (single AZ outage) with automatic failover, whereas a Region-wide catastrophe or regional control-plane outage would still require a Multi-Region architecture
B. Availability Zones are logical folders on a single server, so it provides zero hardware redundancy
C. It protects against a global DNS root server outage, but fails if a single hard drive breaks
D. It protects against application SQL injection bugs, whereas a single VM reboot requires Multi-Region failover
ANSWER: A

A healthcare consortium of five independent hospitals needs a shared cloud infrastructure specifically tailored to meet identical HIPAA/health-data regulatory compliance and clinical auditing requirements, governed jointly by the participating hospitals. Which NIST cloud

deployment model best fits this scenario?
A. Community Cloud
B. Personal Home Lab
C. Single-Departmental Type I Intranet
D. Public Retail Cloud
ANSWER: A

What are the three primary architectural tiers that make up a standard Chef infrastructure environment?
A. Controller, Inventory File, and SSH Agent
B. Chef Workstation, Chef Infra Server, and Chef Nodes
C. Provider, State File, and Backend Bucket
D. Master Node, Pod, and Container Runtime
ANSWER: B

In Chef terminology, what is the fundamental configuration unit that defines a collection of resources (such as packages to install, files to manage, and services to start) and is grouped inside a Cookbook?
A. Playbook
B. Module
C. Recipe
D. Manifest
ANSWER: C

Which programming language serves as the Domain Specific Language (DSL) for writing Chef Recipes and Cookbooks?
A. Ruby
B. Java
C. Python
D. Go
ANSWER: A

Which component running on a managed Chef Node collects system profiling attributes (such as OS platform, IP address, CPU count, memory, and hostname) at the start of every chef-client run and provides them to the node object?
A. Ohai
B. Berkshelf
C. Test Kitchen
D. Knife
ANSWER: A

Which command-line tool executed from the Chef Workstation acts as the primary interface between a local chef-repo and the Chef Infra Server to manage nodes, upload cookbooks, and edit roles or data bags?
A. facter
B. chef-solo
C. knife
D. ohai
ANSWER: C

During the Lab Assignment for configuring Chef to perform infrastructure provisioning, which knife subcommand is used to install the chef-client agent onto a new target node over SSH, register the node with the Chef Server, and perform its initial convergence?
A. knife data bag create
B. knife cookbook upload
C. knife node delete
D. knife bootstrap
ANSWER: D

In a standard Chef Cookbook directory structure, which directory holds dynamically generated configuration files written in Embedded Ruby (.erb) that interpolate node attributes at runtime?
A. files/default/
B. libraries/
C. templates/
D. attributes/
ANSWER: C

A Chef administrator wants to ensure that the httpd package is installed and the httpd service is both enabled at boot and currently running on a Linux node. Which Chef Recipe resource syntax is

valid and idempotent?
A. package 'httpd' { state: 'present' }; service 'httpd' { state: 'started' }
B. yum install -y httpd && systemctl enable --now httpd without any resource block
C. resource 'package' 'httpd' { ensure => installed }
D. package 'httpd' do action :install end; service 'httpd' do action [:enable, :start] end
ANSWER: D

During a chef-client run on a target node, what happens during the 'Compile' phase versus the 'Converge' phase of the two-phase execution model?
A. During Compile, chef-client loads libraries/attributes and evaluates all recipes in the run-list to build an in-memory Resource Collection; during Converge, it iterates through each resource in the collection and executes the provider actions to bring the system to the desired state
B. During Compile, files are deleted from disk; during Converge, cookbooks are uploaded to GitHub
C. During Compile, the Linux kernel is recompiled from source; during Converge, the node reboots into BIOS
D. During Compile, Ohai shuts down the network interface; during Converge, Knife generates SSH keys
ANSWER: A

In Chef attribute precedence, if the same attribute key node['apache']['listen_port'] is defined in multiple places as a default attribute in a Cookbook's attributes/default.rb, a default attribute inside a Recipe, an override attribute inside a Role, and an automatic attribute discovered by Ohai, which level has the highest precedence?
A. The override attribute inside the Role
B. The default attribute in the Cookbook's attributes/default.rb file
C. The default attribute inside the Recipe file
D. The automatic attribute discovered by Ohai
ANSWER: D

In Puppet's standard client-server architecture, what are the names of the central server and the managed client software, and which default TCP port do they use for communication?
A. Puppet Primary Server (Puppet Master) and Puppet Agent over TCP port 8140
B. Puppet Workstation and Puppet Client over TCP port 8080
C. Puppet Controller and Puppet Worker over TCP port 22
D. Puppet Hub and Puppet Spoke over TCP port 9090
ANSWER: A

What file extension is used for Puppet Manifest files where desired system state resources are declared?
A. .pp
B. .yml
C. .tf
D. .rb
ANSWER: A

Which cross-platform system profiling tool is bundled with Puppet to gather facts (such as operating system family, IP address, hostname, and memory size) from the agent node before requesting a catalog?
A. MCollective
B. Ohai
C. Facter
D. Hiera
ANSWER: C

In a Puppet run workflow, what is the 'Catalog' that the Puppet Primary Server sends back to the Puppet Agent?
A. A raw tarball of the Linux kernel source tree
B. An unencrypted text file containing all users' plaintext passwords
C. A compiled JSON document describing the desired state of every resource that should be managed on that specific node, along with dependency ordering
D. A list of commercial software licenses available for purchase
ANSWER: C

During initial Puppet Agent installation and provisioning, how is trust established between a new Puppet Agent node and the Puppet Primary Server?
A. By disabling all firewalls and running Telnet on port 23
B. By logging into the node with an anonymous FTP session
C. By copying /etc/shadow from the agent to the server over TFTP
D. The Puppet Agent generates an SSL/TLS Certificate Signing Request (CSR) and sends it to the

Puppet Server's Certificate Authority (CA), where the administrator signs it using puppetserver ca sign
ANSWER: D

Which Puppet resource declaration syntax correctly ensures that the nginx package is installed and the nginx service is running and enabled at boot?
A. package { 'nginx': ensure => installed } service { 'nginx': ensure => running, enable => true }
B. - name: Install nginx apt: name=nginx state=present
C. resource "package" "nginx" { state = "installed" }
D. package 'nginx' do action :install end
ANSWER: A

Which command is run on a managed node to immediately trigger a manual Puppet Agent run in the foreground with verbose output for testing and troubleshooting?
A. facter --apply
B. puppet apply --noop
C. puppet agent -t (or puppet agent --test)
D. puppetserver reload
ANSWER: C

In Puppet Manifest development, what is the difference between the ordering arrow -> (or require/before) and the notification arrow ~> (or subscribe/notify) between a configuration file resource and a service resource?
A. -> restarts the service every time Puppet runs, whereas ~> deletes the file after starting the service
B. -> enforces execution order only (applying the file before the service), whereas ~> enforces both execution order AND sends a refresh event to restart/reload the service whenever the file resource changes
C. -> works only on Windows agents, whereas ~> works only on Linux agents
D. -> is used for Hiera lookups, whereas ~> is used for Facter custom facts
ANSWER: B

How does Puppet separate site-specific configuration data (such as environment-specific database hostnames or NTP servers) from reusable Puppet Module code?
A. By using Hiera, Puppet's built-in key-value hierarchical data lookup system configured via hiera.yaml
B. By compiling variables into binary .exe files before catalog compilation
C. By hardcoding if/else statements for every server IP address directly inside init.pp
D. By storing all variables inside /etc/fstab on the Puppet workstation
ANSWER: A

During the Lab Assignment for configuring Puppet infrastructure provisioning, an administrator writes a Puppet module named apache. In which exact path inside the module directory must the main apache class (class apache { ... }) be placed so Puppet's autoloader can find it automatically?
A. apache/lib/facter/init.pp
B. apache/manifests/init.pp
C. apache/files/apache.pp
D. apache/templates/main.pp
ANSWER: B

What type of database is at the core of the Prometheus monitoring system, and what primary model does Prometheus use to collect metrics from monitored targets?
A. A Time Series Database (TSDB) that uses an HTTP pull (scrape) model to fetch metrics from /metrics endpoints on configured targets
B. A relational SQL database that waits for targets to push metrics via SMTP
C. A graph database that uses SNMPv1 traps exclusively over UDP port 162
D. A document store that reads log files over SMB file shares
ANSWER: A

What is the default TCP port on which the Prometheus server web UI and HTTP API listen?
A. 8080
B. 9090
C. 9100
D. 9182
ANSWER: B

What is the name of the functional query language used in Prometheus to select, aggregate, and analyze time series data in real time?
A. PromQL (Prometheus Query Language)

B. LogQL
C. SQL (Structured Query Language)
D. GraphQL
ANSWER: A

What are the four core metric types supported by Prometheus client libraries?
A. Integer, Float, String, and Boolean
B. Table, View, Index, and Trigger
C. Event, Alert, Log, and Trace
D. Counter, Gauge, Histogram, and Summary
ANSWER: D

In the Lab Assignment to monitor Linux systems using Prometheus, which agent/exporter is installed on the target Linux host to expose hardware and kernel metrics, and what is its default listening TCP port?
A. Alertmanager on TCP port 9093
B. Blackbox Exporter on TCP port 9115
C. Node Exporter on TCP port 9100
D. cAdvisor on TCP port 8080
ANSWER: C

In the Lab Assignment to monitor Windows systems using Prometheus, which exporter is installed on Windows Server/Client machines to expose WMI and performance counter metrics, and what is its default TCP port?
A. IIS Exporter on TCP port 80
B. SNMP Exporter on TCP port 9116
C. Windows Exporter (formerly WMI Exporter) on TCP port 9182
D. Node Exporter on TCP port 9100
ANSWER: C

In the Prometheus configuration file (prometheus.yml), which top-level section defines the list of target jobs and endpoints (such as localhost:9090 or 192.168.1.50:9100) that Prometheus periodically polls?
A. scrape_configs:
B. remote_write:
C. rule_files:
D. alerting:
ANSWER: A

When using the built-in Prometheus Web UI (http://<server-ip>:9090) to inspect target health, which menu path shows whether each configured exporter endpoint is currently UP or DOWN, along with the last scrape duration and any connection errors?
A. Alerts -> Rules
B. Graph -> Console
C. Help -> Metrics Explorer
D. Status -> Targets
ANSWER: D

How does a Prometheus Counter metric differ from a Gauge metric, and which PromQL function is specifically designed to calculate the per-second average rate of increase of a Counter over a time range vector (such as [5m])?
A. A Counter stores text strings, whereas a Gauge stores histograms; calculated with count_values()
B. A Counter is a cumulative metric that only increases monotonically (or resets to zero on restart, like total HTTP requests or CPU seconds), whereas a Gauge represents a value that can arbitrarily go up or down (like current temperature or free RAM); Counter rate is calculated with rate() or irate()
C. Both Counter and Gauge reset to zero after every scrape interval; calculated with delta()
D. A Counter can go up and down (like memory usage), whereas a Gauge only increases; calculated with sum()
ANSWER: B

An administrator wants to write a PromQL query in the Prometheus Web UI to calculate the current percentage of used RAM on a Linux host monitored by Node Exporter. Given the metrics node_memory_MemTotal_bytes and node_memory_MemAvailable_bytes, which PromQL expression accurately returns the used memory percentage?
A. rate(node_memory_MemTotal_bytes[5m]) / 100
B. node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes
C. (1 - (node_memory_MemAvailable_bytes / node_memory_MemTotal_bytes)) * 100

D. node_memory_MemTotal_bytes + node_memory_MemAvailable_bytes * 100
ANSWER: C

How does Prometheus handle short-lived batch jobs that finish executing before Prometheus's periodic pull scrape interval occurs, and which component handles deduplicating, grouping, silencing, and routing triggered alerts to email, Slack, or PagerDuty?
A. Prometheus opens an SSH session to run the batch job directly, and syslogd sends the emails
B. Short-lived jobs write metrics into /etc/hosts, and promtool sends SMS notifications
C. Short-lived jobs use Node Exporter, and Grafana routes the alerts directly without any other component
D. Short-lived batch jobs push their metrics to an intermediate Prometheus Pushgateway (which Prometheus then scrapes), while Alertmanager handles deduplicating, grouping, silencing, and routing alerts generated by Prometheus alerting rules
ANSWER: D

Before restarting the Prometheus service after editing prometheus.yml and alerting rule files on a Linux server, which command-line utility bundled with Prometheus should be run to validate the syntax of prometheus.yml, and how can Prometheus reload its configuration at runtime without dropping active scrapes?
A. Run systemctl disable prometheus followed by iptables -F
B. Run promtool check config prometheus.yml to verify syntax, then send a SIGHUP signal (kill -HUP <pid>) or an HTTP POST request to /-/reload (when --web.enable-lifecycle is enabled)
C. Run prometheus --repair and reboot the Linux host
D. Run node_exporter --validate and delete the /data TSDB directory
ANSWER: B

In the CALMS framework commonly used to assess an organization's DevOps maturity, what does the acronym CALMS stand for?
A. Culture, Automation, Lean, Measurement, and Sharing
B. Cloud, Containers, Linux, Monitoring, and Security
C. Coding, Auditing, Logging, Merging, and Shipping
D. Configuration, Availability, Latency, Maintenance, and Support
ANSWER: A

In a DevOps CI/CD pipeline, what is the primary distinction between Continuous Delivery and Continuous Deployment?
A. Continuous Delivery compiles code once a year, whereas Continuous Deployment never runs automated tests
B. Continuous Delivery automatically builds, tests, and prepares release-ready artifacts up to staging (requiring a manual approval gate for production deployment), whereas Continuous Deployment automatically deploys every change that passes all automated pipeline tests directly to production without manual intervention
C. Continuous Delivery applies only to database schemas, whereas Continuous Deployment applies only to static HTML files
D. Continuous Delivery is performed before Continuous Integration
ANSWER: B

How do Docker containers differ architecturally from traditional hardware Virtual Machines (VMs)?
A. Docker containers share the host operating system's Linux kernel and isolate application processes in user space, whereas VMs run a separate full guest operating system and kernel on top of a hypervisor
B. Docker containers can only run on Type-1 hypervisors without an operating system
C. Docker containers include a full guest OS kernel and virtualized BIOS per container, whereas VMs share the host kernel
D. Docker containers require at least 16 GB of dedicated RAM per container, whereas VMs require only 4 MB
ANSWER: A

Which two foundational Linux kernel features does the Docker Engine rely on to provide process/network/filesystem isolation and hardware resource usage limits (CPU, RAM, Disk I/O) for containers?
A. SELinux policies and Systemd timers
B. Logical Volume Manager (LVM) and Network File System (NFS)
C. Cron daemons and TCP Wrappers
D. Kernel Namespaces (for isolation) and Control Groups / cgroups (for resource limiting and accounting)
ANSWER: D

In the Docker Engine client-server architecture, what are the three core components and how do

they interact?
A. The Dockerfile, the .git folder, and the SSH daemon
B. The Docker CLI client (docker), the Docker Daemon (dockerd) exposing a REST API over a UNIX socket or network port, and the underlying container runtimes (containerd and runc)
C. The Hypervisor, the vSwitch, and the SAN LUN
D. The Docker GUI, the MySQL database, and the Apache web server
ANSWER: B

What is the relationship between a Docker Image and a Docker Container?
A. A Docker Image can be modified in-place while running without creating a container
B. Each Docker Image can only ever launch a single container at a time on a host
C. A Docker Image is an immutable, layered read-only template containing the application and its dependencies, whereas a Docker Container is a runnable, live instance of an image that adds a thin read-write layer on top
D. A Docker Container is a read-only template, whereas a Docker Image is a running process
ANSWER: C

What is the primary goal of Continuous Integration (CI) in software development?
A. Allowing developers to work on isolated feature branches for six months before merging code
B. Manually copying compiled binaries to production servers via USB drives
C. Having developers frequently merge code changes into a shared mainline repository, where each commit automatically triggers an automated build and unit/integration test suite to detect integration defects early
D. Eliminating version control systems in favor of shared network folders
ANSWER: C

How does Docker's Union File System (such as OverlayFS / overlay2) handle a situation where a running container modifies an existing file that originally resided in one of the underlying read-only image layers?
A. It uses a Copy-on-Write (CoW) strategy: the file is copied up from the read-only image layer (lowerdir) into the container's thin writable layer (upperdir), and the modifications are applied only to that upper copy
B. It throws a read-only filesystem kernel panic unless the container is run with --privileged
C. It duplicates the entire multi-gigabyte base image into RAM before changing a single byte
D. It overwrites the file directly inside the base read-only image layer, mutating all other containers running from that image
ANSWER: A

Which specific Linux kernel namespaces are created by default for a new Docker container to isolate its process IDs, network stack, mount points, inter-process communication, and hostname?
A. tcp, udp, icmp, arp, and bgp namespaces
B. uid, gid, acl, pam, and sudo namespaces
C. pid (Process ID), net (Networking), mnt (Mount points), ipc (Inter-Process Communication), and uts (UNIX Time-Sharing / Hostname)
D. ext4, xfs, btrfs, zfs, and ntfs namespaces
ANSWER: C

In DevOps telemetry and DORA (DevOps Research and Assessment) performance measurement, which four key metrics evaluate software delivery throughput and operational stability?
A. Cost per Click, Monthly Active Users, Page Load Speed, and Bounce Rate
B. Lines of Code Written, Number of Meetings, CPU Clock Speed, and Disk Capacity
C. Total Defects Logged, QA Headcount, Server Uptime Only, and Ticket Backlog Size
D. Deployment Frequency, Lead Time for Changes, Change Failure Rate, and Time to Restore Service (MTTR)
ANSWER: D

Which Docker CLI command lists all containers on the host, including both currently running containers and stopped/exited containers?
A. docker ps
B. docker inspect
C. docker images
D. docker ps -a (or docker container ls -a)
ANSWER: D

Which Dockerfile instruction must be the first non-comment instruction in a standard Dockerfile (after any optional global ARG) to specify the base image for the build?
A. RUN
B. CMD
C. COPY

D. FROM
ANSWER: D

Which Docker command downloads a pre-built image from a container registry (such as Docker Hub) to the local host without starting a container?
A. docker commit
B. docker run
C. docker pull
D. docker push
ANSWER: C

In a Dockerfile, what is the critical difference between the RUN instruction and the CMD instruction?
A. RUN can only execute shell scripts, whereas CMD can only install apt/yum packages
B. There is no difference; RUN and CMD are interchangeable aliases
C. RUN executes commands during the docker build phase and commits the result as a new image layer, whereas CMD defines the default command and/or arguments executed at runtime when a container is launched from the built image
D. RUN executes when a container starts up, whereas CMD executes during image build
ANSWER: C

In a Dockerfile, why do Docker best practices recommend using the COPY instruction over ADD when simply copying local files or executables from the build context into the image?
A. Because ADD was deprecated and removed in Docker Engine v1.12
B. Because COPY compresses binary files by 90% automatically
C. Because COPY automatically downloads remote URLs and extracts encrypted ZIP files
D. Because COPY is explicit and transparent (only copying local files/directories), whereas ADD has implicit side effects such as automatically extracting local compressed tar archives and fetching remote URLs
ANSWER: D

In the Lab Assignment where you compile a C program into an executable binary (app) on the base Linux machine and build a Docker image named name:v1 from the current directory's Dockerfile, which docker build command correctly builds and tags the image?
A. docker build -n Name:v1 Dockerfile
B. docker run --build name:v1 .
C. docker image create name:v1
D. docker build -t name:v1 .
ANSWER: D

In the Lab Assignment where you run a containerized C program that requires interactive user input from stdin (such as scanf or fgets), which flags must be passed to docker run so the user can type input interactively into the container terminal?
A. docker run -it name:v1 (combining -i / --interactive to keep STDIN open and -t / --tty to allocate a pseudo-TTY)
B. docker run -d -P name:v1
C. docker run --read-only name:v1
D. docker run --rm --detach name:v1
ANSWER: A

A student compiles a C program (gcc prog.c -o prog) on an Ubuntu 24.04 host (which dynamically links against glibc 2.39), copies the prog binary into an alpine:latest container (which uses musl libc), and runs the container. The container immediately fails with exec ./prog: no such file or directory. How can this issue be resolved cleanly?
A. Rename the binary from prog to prog.exe
B. Either compile the C program inside a multi-stage Dockerfile using the matching target base environment (or compile statically with gcc -static prog.c -o prog), or use a compatible glibc-based runtime image
C. Expose TCP port 80 in the Dockerfile using EXPOSE 80
D. Run docker system prune on the host machine
ANSWER: B

How do ENTRYPOINT and CMD interact when both are specified in exec form (["..."]) inside the same Dockerfile?
A. CMD completely overrides and erases ENTRYPOINT during docker build
B. The container fails to build because a Dockerfile cannot contain both ENTRYPOINT and CMD
C. ENTRYPOINT defines the fixed executable that always runs when the container starts, while CMD provides default arguments passed to that ENTRYPOINT (which can be easily overridden by passing arguments at the end of docker run)

D. ENTRYPOINT runs before FROM, whereas CMD runs after EXPOSE
ANSWER: C

When running a live troubleshooting session against a running background container named app_server, what is the behavioral difference between docker attach app_server and docker exec -it app_server /bin/bash?
A. docker attach copies files into the container, whereas docker exec exports logs
B. docker attach connects the terminal directly to the container's primary PID 1 process (meaning pressing Ctrl+C can terminate PID 1 and stop the container), whereas docker exec -it spawns a separate secondary process inside the container's namespaces without interrupting PID 1 when exiting
C. docker attach starts a brand-new bash process, whereas docker exec connects to PID 1
D. docker exec only works on stopped containers, whereas docker attach works on running containers
ANSWER: B

When you launch a standalone Docker container without specifying a --network flag, which default network driver does Docker attach the container to?
A. bridge
B. host
C. overlay
D. macvlan
ANSWER: A

Why are Docker Volumes or Bind Mounts required when running stateful workloads (such as databases or persistent file stores) in Docker containers?
A. Because Docker volumes increase CPU clock frequency by 50%
B. Because Docker images cannot be pulled from Docker Hub unless a volume is created first
C. Because a container's writable layer is ephemeral and tied to the container's lifecycle, meaning all data written inside the container is lost when the container is removed (docker rm)
D. Because containers cannot read files larger than 1 MB without a volume
ANSWER: C

In the Lab Assignment: 'Create an httpd container running on port 8000. Name the container as web5. Do not map any directory', which command properly starts this container in detached mode?
A. docker run -d --name web5 -p 8000:80 httpd
B. docker create --port 8000 web5 httpd
C. docker run -d --name web5 -p 80:8000 httpd
D. docker run -d --name web5 -v /var/www/html httpd:8000
ANSWER: A

In the web5 (httpd) Lab Assignment, since no host directory was mapped (-v was not used), which Docker command copies your local index.html file into the default document root of the running httpd (web5) container so the webpage updates immediately?
A. docker cp index.html web5:/var/www/html/index.html
B. docker push index.html web5:/etc/httpd/index.html
C. docker import index.html web5:/root/index.html
D. docker cp index.html web5:/usr/local/apache2/htdocs/index.html
ANSWER: D

In the web5 Lab Assignment, after modifying index.html on the host machine and running docker cp index.html web5:/usr/local/apache2/htdocs/ a second time, why does refreshing http://localhost:8000 immediately show the updated webpage without restarting the web5 container, and what happens if docker rm -f web5 is later executed?
A. Apache serves static files dynamically from its document root in the container's writable layer on each HTTP request, so the updated file is served immediately; however, because no volume or bind mount was used, removing web5 permanently deletes the modified index.html inside the container
B. Docker automatically commits the modified index.html back into the official httpd base image on Docker Hub
C. docker cp creates a permanent NFS mount between the host and container that survives docker rm
D. The website will not update until the entire Linux host machine is rebooted
ANSWER: A

What is the key architectural difference between a Docker Named Volume (-v mydata:/var/lib/mysql) and a Bind Mount (-v /home/user/data:/var/lib/mysql)?
A. Bind Mounts can be managed using docker volume create, whereas Named Volumes cannot
B. Named Volumes work only on Windows hosts, whereas Bind Mounts work only on Linux hosts
C. Named Volumes are fully managed by Docker inside Docker's storage directory (/var/lib/docker/volumes/) and isolated from host path structure, whereas Bind Mounts map an

explicit, user-specified file or directory path from the host filesystem directly into the container
D. Named Volumes are stored only in volatile RAM, whereas Bind Mounts are stored on disk
ANSWER: C

Which Docker network driver removes network isolation between the container and the Docker host, causing the container to share the host's network namespace and bind directly to the host's IP address and ports?
A. none
B. overlay
C. bridge
D. host
ANSWER: D

Two containers, api and db, are started on the same host. When attached to the default bridge network (docker0), api cannot ping db by its container name (ping db), but when both are attached to a user-defined bridge network (docker network create app-net), ping db succeeds automatically. Why?
A. User-defined bridge networks provide built-in embedded Docker DNS resolution (127.0.0.11) that automatically resolves container names and aliases to container IP addresses, whereas the legacy default bridge network only supports IP-based communication (unless deprecated --link flags are used)
B. User-defined bridge networks automatically install BIND9 inside every container image
C. The default bridge network uses IPv6 exclusively, whereas user-defined networks use IPv4
D. The default bridge network blocks all ICMP and TCP traffic between containers by default
ANSWER: A

Which Docker network driver is used to enable multi-host container-to-container communication across multiple physical Docker daemon nodes (such as in a Docker Swarm cluster) using VXLAN encapsulation, and which driver assigns a unique MAC address to each container so it appears as a physical device directly on the underlying physical LAN?
A. none for multi-host VXLAN; ipvlan for disabling Ethernet interfaces
B. overlay for multi-host VXLAN communication; macvlan for assigning unique MAC addresses on the physical LAN
C. nat for multi-host VXLAN; loopback for unique MAC addresses
D. bridge for multi-host VXLAN; host for unique MAC addresses on the physical LAN
ANSWER: B

When a new, empty Docker Named Volume is mounted into a container directory that already contains pre-existing files in the container image (for example, mounting a brand-new named volume apache-htdocs to /usr/local/apache2/htdocs on an httpd container), how does Docker handle those pre-existing image files compared to mounting an empty host directory via a Bind Mount?
A. Docker throws a mount conflict error and refuses to start the container in both cases
B. Both the Named Volume and the Bind Mount delete all files in the container directory
C. The Bind Mount copies the image files to the host, whereas the Named Volume formats the directory as read-only
D. Docker automatically copies the pre-existing files from the image directory into the newly initialized empty Named Volume, whereas an empty host Bind Mount obscures/hides the underlying image directory's files without copying them
ANSWER: D

What is the primary purpose of Docker Compose, and what is the default filename it looks for in the current working directory?
A. To compile Linux kernels; looks for Makefile
B. To provision AWS VPC routers; looks for main.tf
C. To define and run multi-container Docker applications using a declarative YAML file; looks for compose.yaml or docker-compose.yml
D. To monitor Windows event logs; looks for prometheus.yml
ANSWER: C

Which Docker Compose command builds (if needed), creates, starts, and attaches to all containers defined in docker-compose.yml in the background (detached mode)?
A. docker compose start -b
B. docker compose run --all
C. docker compose init
D. docker compose up -d (or docker-compose up -d)
ANSWER: D

Before you can push a locally built Docker image to your personal or organizational repository on

Docker Hub, which command must you execute to authenticate your Docker CLI session?
A. docker register
B. docker connect
C. docker login
D. docker auth init
ANSWER: C

You have a local Docker image named name:v1 and your Docker Hub username is cdacuser. Which two commands are required to properly tag this local image for your Docker Hub repository myrepo and upload it to Docker Hub?
A. docker export name:v1 > cdacuser.tar followed by docker import cdacuser.tar
B. docker rename name:v1 cdacuser/myrepo:v1 followed by docker upload cdacuser/myrepo:v1
C. docker commit name:v1 docker.io/myrepo followed by docker send myrepo
D. docker tag name:v1 cdacuser/myrepo:v1 followed by docker push cdacuser/myrepo:v1
ANSWER: D

In a docker-compose.yml file, what are the three primary top-level keys used to define the application's containers, persistent storage, and custom communication channels?
A. containers:, disks:, and switches:
B. tasks:, handlers:, and vars:
C. pods:, pvcs:, and ingresses:
D. services:, volumes:, and networks:
ANSWER: D

What happens when you execute docker compose down -v in a project directory compared to running docker compose stop?
A. docker compose down -v stops and removes the containers, default networks, AND removes the named volumes declared in the volumes section of the Compose file, whereas docker compose stop only halts the running container processes without removing containers, networks, or volumes
B. docker compose down -v prints the version number of Docker Compose and exits
C. docker compose down -v upgrades all running containers to the latest image tag
D. docker compose stop deletes all images and volumes, whereas docker compose down -v only pauses containers
ANSWER: A

When a user runs docker pull nginx without specifying a registry hostname or tag, which default registry hostname, namespace, and image tag does Docker automatically resolve?
A. localhost:5000/default/nginx:master
B. gcr.io/google/nginx:stable
C. docker.io/library/nginx:latest
D. quay.io/redhat/nginx:v1
ANSWER: C

In a docker-compose.yml file, a web service includes depends_on: [db]. Why might the web application still fail to connect to the PostgreSQL db service immediately upon startup, and how can depends_on be configured in Compose to wait until PostgreSQL is genuinely ready to accept connections?
A. Because depends_on only works during docker compose down
B. Because depends_on starts containers in alphabetical order; fixed by renaming db to aaa_db
C. Because Docker Compose does not allow two services to communicate over TCP; fixed by using network_mode: none
D. By default, depends_on: [db] only waits for the db container to start (service_started), not for the database engine inside the container to finish initializing; this is solved by defining a healthcheck (e.g., pg_isready) on the db service and setting depends_on: db: condition: service_healthy
ANSWER: D

In a docker-compose.yml service definition, what is the difference between the ports: directive (e.g., ports: ["8080:80"]) and the expose: directive (e.g., expose: ["3306"])?
A. expose: publishes the port to all public interfaces on the host, whereas ports: restricts it to 127.0.0.1
B. Both directives perform the exact same host port binding
C. ports: encrypts traffic with TLS, whereas expose: uses unencrypted UDP
D. ports: publishes and binds the container port to a port on the host machine (making it accessible outside the host), whereas expose: only documents/exposes the port to other linked services on the same internal Docker Compose network without publishing it to the host machine
ANSWER: D

When an image tag such as myorg/webapp:v2.1 is updated and pushed again to Docker Hub with new

code changes, why is referencing an image by its sha256 digest (myorg/webapp@sha256:4a8b...) preferred in strict production deployments over referencing a mutable tag like :latest or :v2.1?
A. Because image tags are mutable pointers that can be overwritten in a registry to point to a different image build, whereas an image's content-addressable sha256 digest is cryptographically immutable and guarantees the exact same binary image layers on every node
B. Because Docker Hub deletes tags after 7 days unless a digest is used
C. Because docker tag cannot create tags that contain numbers or dots
D. Because sha256 digests compress the image size by 50% during docker pull
ANSWER: A

What type of Version Control System (VCS) is Git?
A. A Relational Database Management System for SQL tables
B. A Centralized Version Control System (CVCS) that requires a constant connection to a central server for every commit
C. A Distributed Version Control System (DVCS) where every developer's working copy contains the full repository history
D. A Local File Locking System that stores only the latest version of each file
ANSWER: C

What are the three main local states (and corresponding areas) that files move through in a standard Git project?
A. Public, Private, and Protected
B. Uploaded, Downloaded, and Archived
C. Compiled, Linked, and Executed
D. Modified (Working Directory), Staged (Staging Area / Index), and Committed (.git Repository Directory)
ANSWER: D

In the Lab Assignment for Git Configuration, which two commands configure your user name and email address globally for all Git repositories on your local computer?
A. git init --user "Name" and git init --email "user@example.com"
B. git login --username "Name" and git login --email "user@example.com"
C. git set user.name "Name" and git set user.email "user@example.com"
D. git config --global user.name "Name" and git config --global user.email "user@example.com"
ANSWER: D

When setting up a brand-new local project folder from the command line, which Git command initializes an empty Git repository by creating the hidden .git subdirectory?
A. git init
B. git create
C. git new
D. git start
ANSWER: A

How does Git internally store and identify committed content in its object database (.git/objects)?
A. It stores data as a series of point-in-time snapshots (using blob, tree, commit, and tag objects) where every object is checksummed and addressed by its 40-character SHA-1 (or SHA-256) cryptographic hash
B. It stores only line-by-line forward text diffs indexed by sequential integer IDs (1, 2, 3...)
C. It compresses each branch into a Microsoft Access .mdb file
D. It uploads every file to GitHub before generating a timestamp ID
ANSWER: A

Which Git command displays the current state of the working directory and the staging area, showing which files are untracked, modified but unstaged, or staged and ready to be committed?
A. git show
B. git log
C. git diff --cached
D. git status
ANSWER: D

In Git configuration, what are the three configuration scope flags, and where is the --global configuration stored on a Linux system?
A. --system (all users on host), --global (current user), and --local (current repository); --global is stored in ~/.gitconfig (or ~/.config/git/config)
B. --master, --main, and --branch; stored in .gitignore
C. --root, --home, and --dir; stored in /var/log/git.conf
D. --public, --private, and --secret; stored in /etc/passwd

ANSWER: A

A developer modifies app.py and runs git add app.py. Immediately afterward, the developer makes a second edit to app.py in the working directory without running git add again. Which commands will inspect (1) the unstaged changes between the working directory and the staging area, and (2) the staged changes that will go into the next commit?
A. git status for unstaged changes, and git log for staged changes
B. git show for unstaged changes, and git blame for staged changes
C. git checkout for unstaged changes, and git reset for staged changes
D. git diff for unstaged changes (Working Directory vs. Index), and git diff --staged (or git diff --cached) for staged changes (Index vs. HEAD)
ANSWER: D

In Git's internal architecture, what is HEAD, and what does it mean when a repository is in a 'detached HEAD' state?
A. HEAD is the first initial commit ever created in the repository; 'detached HEAD' means the .git folder was deleted
B. HEAD is a symbolic reference pointer normally pointing to the current branch reference (e.g., refs/heads/main), which in turn points to the latest commit; 'detached HEAD' occurs when HEAD points directly to a specific commit SHA rather than a named branch
C. HEAD is the staging index file; 'detached HEAD' means merge conflicts exist in the working tree
D. HEAD is the remote server URL; 'detached HEAD' means the computer is disconnected from the internet
ANSWER: B

If user.email is set to corp@company.com using git config --system, set to personal@gmail.com using git config --global, and set to project@cdac.in using git config --local inside a specific repository, which email address will Git attach to a new commit created inside that repository?
A. corp@company.com because --system overrides all user settings
B. project@cdac.in because Git resolves configuration precedence from most specific to least specific (--local overrides --global, which overrides --system)
C. personal@gmail.com because --global overrides --local
D. Git will refuse to commit until two of the three configuration files are deleted
ANSWER: B

Which Git command copies an existing remote repository from GitHub to your local computer, automatically creating a working directory, initializing .git, fetching all history, and setting up the default remote named origin?
A. git clone https://github.com/user/repo.git
B. git fork https://github.com/user/repo.git
C. git init https://github.com/user/repo.git
D. git pull --new https://github.com/user/repo.git
ANSWER: A

After staging your new project files with git add ., which command records the snapshot permanently in your local Git repository with the commit message 'Initial commit'?
A. git stage -m "Initial commit"
B. git commit -m "Initial commit"
C. git save -m "Initial commit"
D. git push -m "Initial commit"
ANSWER: B

Which command uploads your committed local changes on the main branch to the remote repository named origin on GitHub and sets the upstream tracking reference?
A. git commit --remote origin main
B. git push -u origin main
C. git send origin/main
D. git upload origin main
ANSWER: B

You created a local Git repository first using git init and now want to link it to a newly created empty repository on GitHub (https://www.google.com/search?q=https://github.com/cdac/devops.git). Which command adds this GitHub URL as the remote named origin?
A. git config remote.origin https://www.google.com/search?q=https://github.com/cdac/devops.git
B. git clone --remote origin https://www.google.com/search?q=https://github.com/cdac/devops.git
C. git remote add origin https://www.google.com/search?q=https://github.com/cdac/devops.git
D. git remote set-url origin https://www.google.com/search?q=https://github.com/cdac/devops.git
ANSWER: C

What is the difference between git fetch origin and git pull origin main?
A. git fetch deletes local uncommitted changes, whereas git pull only checks network latency
B. git fetch uploads local commits to GitHub, whereas git pull downloads commits from GitHub
C. git fetch downloads new commits and branch references from the remote repository into your local remote-tracking branches (e.g., origin/main) without modifying your current working directory or merging, whereas git pull performs a git fetch immediately followed by a git merge (or git rebase) into your current local branch
D. There is no difference; git fetch and git pull are identical aliases
ANSWER: C

Which special file placed in a Git project folder specifies intentionally untracked files and directories (such as compiled binaries, .env secrets, node_modules/, or .terraform/) that Git should not track or stage?
A. .gitconfig
B. .gitattributes
C. .gitmodules
D. .gitignore
ANSWER: D

On GitHub, what is the purpose of creating a 'Fork' and submitting a 'Pull Request (PR)' in an open-source or team workflow?
A. A Fork is identical to git branch, and a Pull Request is identical to git clone
B. A Fork creates a personal server-side copy of another user's or organization's repository under your GitHub account, and a Pull Request proposes merging your committed branch changes back into the upstream base repository with code review and CI checks
C. A Pull Request forces all developers to download a ZIP archive of the project
D. A Fork deletes the original repository, and a Pull Request restores it from backup
ANSWER: B

A developer attempts to run git push origin main, but Git rejects the push with the error: ! [rejected] main -> main (non-fast-forward) Updates were rejected because the remote contains work that you do not have locally. What caused this error, and what is the safe standard resolution?
A. The developer's hard drive is full; resolve by running git gc --prune=now
B. Another collaborator pushed new commits to origin/main after the developer last synced, causing the local and remote histories to diverge; resolve by running git pull origin main (or git pull --rebase origin main), resolving any merge conflicts locally, and then running git push origin main
C. The .git directory is corrupted; resolve by deleting the local folder and rewriting the code from scratch
D. GitHub does not allow more than 10 commits per branch; resolve by creating a new repository
ANSWER: B

When pushing changes to GitHub over HTTPS or SSH from the command line, GitHub no longer accepts routine account login passwords for Git operations. Which two secure authentication mechanisms are supported for CLI git push and git clone operations?
A. Passing the account password in the URL query string (?password=...) over HTTP port 80
B. Sending an SNMP community string to https://www.google.com/search?q=api.github.com
C. Telnet plaintext login or anonymous FTP
D. A Personal Access Token (PAT / OAuth token) used over HTTPS, or an SSH public/private key pair registered in the user's GitHub account settings used over an SSH remote URL (git@github.com:user/repo.git)
ANSWER: D

A developer already committed a credential file (config.secret) in a previous commit, and only afterward adds config.secret to .gitignore. Why does Git continue to track modifications to config.secret, and which command removes config.secret from Git's tracking index without deleting the physical file from the developer's local working directory?
A. .gitignore requires a system reboot to take effect; run git clean -fdx
B. Git ignores .gitignore unless the file is executable (chmod +x .gitignore)
C. Run git reset --hard HEAD~1 which deletes all project files permanently
D. .gitignore only applies to untracked files; run git rm --cached config.secret followed by a new commit to stop tracking it while keeping the local file on disk
ANSWER: D

What is Jenkins primarily used for in a DevOps toolchain, and what runtime environment is required on a server to run Jenkins?
A. An open-source automation server used for Continuous Integration and Continuous Delivery (CI/CD) that runs on the Java Virtual Machine (JVM / Java JDK)
B. A container hypervisor written in C++ that requires a Windows Server kernel
C. A configuration management database written in Ruby that requires PostgreSQL

D. A cloud provider virtual network router that requires Python 2.7
ANSWER: A

What is the default HTTP TCP port on which a freshly installed Jenkins server listens, and where is the initial administrator unlock password stored on a Linux installation?
A. Port 50000; stored in /var/log/jenkins/init.log
B. Port 9090; stored in /etc/jenkins/passwd
C. Port 8443; stored in /root/.jenkins_secret
D. Port 8080; stored in /var/lib/jenkins/secrets/initialAdminPassword
ANSWER: D

In Jenkins Pipeline-as-Code, what is the default name of the text file committed to a project's source control repository that defines the entire CI/CD pipeline?
A. pipeline.yaml
B. Makefile
C. build.xml
D. Jenkinsfile
ANSWER: D

On a Linux server where Jenkins is installed via standard system packages, which directory serves as the default JENKINS_HOME storing all job configurations, build logs, plugins, and workspace artifacts?
A. /var/lib/jenkins
B. /usr/local/jenkins
C. /etc/default/jenkins
D. /opt/jenkins
ANSWER: A

How does a Jenkins 'Freestyle Project' differ from a Jenkins 'Pipeline' job?
A. A Freestyle project is configured primarily through web UI forms with sequential build steps, whereas a Pipeline job defines complex, multi-stage CI/CD workflows (supporting parallelism, conditionals, approvals, and version control) as code in a Jenkinsfile
B. Freestyle projects can only compile Java code, whereas Pipeline jobs can only compile C code
C. Freestyle projects run exclusively on Kubernetes, whereas Pipeline jobs cannot use plugins
D. Freestyle projects do not generate console output logs
ANSWER: A

What are the two syntaxes supported for writing a Jenkinsfile in Jenkins Pipeline, and which top-level block begins a Declarative Pipeline?
A. Bash and PowerShell; begins with #!/bin/jenkins
B. JSON and XML; begins with <pipeline>
C. YAML and TOML; begins with workflow:
D. Declarative Pipeline and Scripted Pipeline (Groovy-based DSL); a Declarative Pipeline must begin with the top-level pipeline { ... } block
ANSWER: D

Within a Jenkins Declarative Pipeline (pipeline { ... }), which four core sections are used to specify where the pipeline runs, the logical phases of the delivery process, the actual commands executed in each phase, and actions taken based on build completion status?
A. nodes, tasks, commands, and handlers
B. runners, jobs, scripts, and webhooks
C. agent, stages (containing stage), steps, and post
D. hosts, vars, roles, and outputs
ANSWER: C

In a Jenkins job configuration, a DevOps engineer sets the Build Trigger to H/15 * * * * under 'Poll SCM', while another job sets H/15 * * * * under 'Build Periodically'. What is the operational difference between these two configurations?
A. 'Poll SCM' queries the version control repository roughly every 15 minutes and triggers a build ONLY if new commits are detected since the last build, whereas 'Build Periodically' triggers a build every 15 minutes unconditionally even if no code has changed
B. 'Poll SCM' requires a webhook from GitHub, whereas 'Build Periodically' deletes the workspace every 15 hours
C. The H symbol stands for Hours, meaning both run once every 15 hours
D. Both configurations force a full build every 15 seconds regardless of source code changes
ANSWER: A

In a distributed Jenkins master/agent (controller/agent) architecture, why is it a critical security and stability best practice to set the number of executors on the built-in Jenkins

Controller node to 0 and execute all builds on dedicated Jenkins Agent nodes (or ephemeral Docker/Kubernetes agents)?
A. Because Jenkins Controller only supports Windows agents over RDP
B. Because build scripts executing on the built-in Controller node run with access to the JENKINS_HOME filesystem (including encrypted credentials, secrets, and master configuration) and can exhaust CPU/memory needed by the Jenkins web UI and scheduler
C. Because setting executors to 0 speeds up Java garbage collection by 10x
D. Because the Jenkins Controller cannot install Git or Docker on Linux
ANSWER: B

Inside the post { ... } block of a Declarative Jenkinsfile, an engineer wants to (1) always clean up temporary workspace containers regardless of outcome, (2) send a Slack notification only when a previously failing pipeline turns green, and (3) page the on-call engineer only when the build fails. Which three post condition blocks correspond to (1), (2), and (3)?
A. cleanup, changed, and aborted
B. default, repaired, and crashed
C. finally, success, and error
D. always, fixed, and failure
ANSWER: D

When integrating a GitHub repository with Jenkins so that a pipeline triggers immediately whenever a developer pushes a commit (without waiting for Poll SCM intervals), what mechanism is configured in the GitHub repository settings?
A. An FTP cron job
B. An SNMP trap to port 161
C. A GitHub Webhook sending an HTTP POST payload to <jenkins-url>/github-webhook/
D. An SSH port-forwarding tunnel
ANSWER: C

On a Linux server where both Jenkins and Docker are installed, which Linux group must the jenkins system user be added to so that Jenkins pipeline stages can execute docker build and docker run commands without sudo permission errors?
A. wheel
B. adm
C. systemd-journal
D. docker
ANSWER: D

In the Session 18 & 19 Lab Assignment: 'Create a C program to take input from the user and test it. Then create an image using Dockerfile... Then also export the image in a cprog-back.tar file', which Docker CLI command saves a Docker image (including all its layers, tags, and metadata) into the archive cprog-back.tar?
A. docker logs <image-name> > cprog-back.tar
B. docker save -o cprog-back.tar <image-name>
C. docker volume export cprog-back.tar
D. docker cp <image-name> cprog-back.tar
ANSWER: B

Regarding the cprog-back.tar Lab Assignment, what is the technical difference between docker save -o cprog-back.tar <image> (reloaded via docker load -i) and docker export -o cprog-back.tar <container> (reloaded via docker import)?
A. docker save works on containers and discards history, whereas docker export works on images and preserves layers
B. docker save archives a Docker image preserving all layer history, tags, and CMD/ENTRYPOINT metadata, whereas docker export exports a flattened filesystem snapshot of a container instance without preserving image layer history or Dockerfile metadata
C. docker save uploads the .tar file directly to Docker Hub, whereas docker export uploads to GitHub
D. Both commands produce byte-for-byte identical tar archives
ANSWER: B

When a Jenkins pipeline needs to authenticate to Docker Hub to push a newly built image (docker push), how should the Docker Hub username and password/token be stored and accessed in the Jenkinsfile?
A. Hardcode the plaintext password directly inside the Jenkinsfile and push it to a public GitHub repo
B. Store the password in /tmp/pass.txt with chmod 777 permissions
C. Store the credentials in the Jenkins Credentials Manager (as 'Username with password' or 'Secret text') and inject them safely at runtime using the

withCredentials([usernamePassword(...)]) block or docker.withRegistry
D. Prompt the developer to type the password into the Jenkins server physical keyboard during every build
ANSWER: C

In the Lab Assignments to automate web deployment using Jenkins, Docker, and GitHub with the nginx image versus the httpd image, what are the default container document root directories where static web files (index.html) must be copied inside an official nginx image versus an official httpd image?
A. /opt/web for both nginx and httpd
B. /usr/share/nginx/html for nginx and /usr/local/apache2/htdocs for httpd
C. /var/www/nginx for nginx and /var/www/apache for httpd
D. /etc/nginx/html for nginx and /etc/httpd/htdocs for httpd
ANSWER: B

In a multi-stage Jenkins CI/CD pipeline that automates building and deploying an nginx or httpd web container from GitHub, what is the logical order of stages inside the Jenkinsfile?
A. Checkout SCM (GitHub) -> Build Docker Image (docker build) -> Test Container -> Push Image to Registry (docker push) -> Deploy/Update Running Web Container
B. Deploy Container -> Push to Docker Hub -> Build Docker Image -> Checkout SCM
C. Delete GitHub Repo -> Install Java -> Format Disk -> Build Image
D. Push to Docker Hub -> Write Dockerfile -> Clone Repository
ANSWER: A

During the automated Deploy stage of the Jenkins CI/CD pipeline for the nginx or httpd web container, the first build succeeds with docker run -d --name web-prod -p 8085:80 myuser/nginx-web:latest, but the second build fails with Conflict. The container name "/web-prod" is already in use. How should the Deploy stage script be written so every subsequent pipeline build cleanly replaces the running container?
A. Run docker system prune -a while the container is still running without stopping it
B. Stop and remove any existing container with that name before launching the new container (for example: docker rm -f web-prod || true followed by docker run -d --name web-prod -p 8085:80 myuser/nginx-web:${BUILD_NUMBER})
C. Change the host port manually in the Jenkins UI before every single commit
D. Reinstall the Docker daemon during every pipeline execution
ANSWER: B

In the Lab Assignment where a C program takes input from stdin, why will a Jenkins pipeline stage hang indefinitely if it runs the C program container without providing input, and how can automated testing of a stdin-driven C program container be executed non-interactively inside a Jenkins pipeline step?
A. It hangs because the C program blocks waiting for EOF or newline on stdin; solved in an automated CI step by piping test input into the container with the -i flag only (without -t), such as echo "42" | docker run --rm -i cprog:v1
B. It hangs because GCC strips all scanf calls inside containers
C. It hangs because Jenkins cannot run C binaries; solved by rewriting the program in HTML
D. It hangs because Docker containers do not support stdout; solved by passing -d
ANSWER: A

In a containerized Jenkins setup where Jenkins itself runs inside a Docker container and needs to build Docker images during pipeline runs, why do security architects warn against mounting the host's /var/run/docker.sock directly into the Jenkins container without isolation, and what is the implication of having write access to /var/run/docker.sock?
A. Mounting /var/run/docker.sock disables network access inside the Jenkins container
B. /var/run/docker.sock only supports pulling images and cannot build Dockerfiles
C. Mounting /var/run/docker.sock corrupts the JENKINS_HOME/config.xml file on restart
D. Write access to /var/run/docker.sock grants root-equivalent privileges over the underlying Docker host (allowing any pipeline script to launch a privileged container mounting the host root filesystem -v /:/host) rather than isolating builds
ANSWER: D

In Amazon Web Services (AWS) global infrastructure terminology, what is the relationship between an AWS Region and an Availability Zone (AZ)?
A. An AWS Region is a distinct geographic area around the world consisting of multiple isolated, physically separated Availability Zones (each comprising one or more discrete data centers with redundant power, networking, and connectivity)
B. An Availability Zone is a country, and a Region is a single server rack inside that country
C. Each AWS Region contains exactly one Availability Zone located in a single building
D. Availability Zones are CDN edge caching routers, whereas Regions are S3 buckets

ANSWER: A

Which three AWS services correspond respectively to (1) resizable virtual servers in the cloud, (2) serverless event-driven compute functions, and (3) scalable object storage?
A. Amazon CloudFront, AWS Glue, and Amazon EBS
B. AWS RDS, AWS ECS, and AWS EFS
C. Amazon VPC, AWS IAM, and Amazon Route 53
D. Amazon EC2, AWS Lambda, and Amazon S3
ANSWER: D

When launching a new Amazon EC2 instance in the Lab Assignment, what does 'AMI' stand for, and what is its purpose?
A. Amazon Machine Image; provides the pre-configured template (operating system, root volume structure, and optional software) required to launch an EC2 instance
B. Access Management Identity; generates SSH key pairs
C. Amazon Memory Interface; specifies the RAM speed of the hypervisor
D. Automated Monitoring Instance; collects CloudWatch billing alarms
ANSWER: A

When creating an Amazon S3 bucket in the Lab Assignment, which requirement applies to the S3 bucket name, and what is the default public access state of a newly created S3 bucket?
A. Bucket names only need to be unique within a single VPC, and all new buckets are public by default
B. Bucket names can contain uppercase letters, spaces, and underscores, and require an EC2 instance to exist first
C. S3 buckets are block-storage volumes that must be formatted with mkfs.ext4 before uploading files
D. S3 bucket names must be globally unique across all AWS accounts and partitions (DNS-compliant), and 'Block all public access' is enabled by default on newly created buckets
ANSWER: D

What is the maximum execution timeout duration allowed for a single invocation of an AWS Lambda function, and how is AWS Lambda billed?
A. Unlimited execution time; billed by the number of lines of source code
B. Maximum timeout is 24 hours; billed as a flat monthly fee per VPC
C. Maximum timeout is 15 minutes (900 seconds); billed based on the number of requests (invocations) and the compute duration (GB-seconds of memory allocated multiplied by execution time in milliseconds)
D. Maximum timeout is 30 seconds; billed only when the function throws an unhandled exception
ANSWER: C

When configuring a Security Group for a new Linux EC2 web server instance so that an administrator can SSH into it and users can browse its HTTP website, which inbound ports must be allowed, and how do Security Groups handle return (outbound response) traffic?
A. Allow UDP port 53 and ICMP; Security Groups are stateless so both Inbound and Outbound rules must match ephemeral ports
B. Allow TCP port 3389 (RDP) only; Security Groups operate at Layer 2 using MAC addresses
C. Allow TCP port 22 (SSH) and TCP port 80 (HTTP); because AWS Security Groups are stateful, return traffic for allowed inbound connections is automatically permitted regardless of outbound rules
D. Allow TCP port 21 and TCP port 8080; return traffic must be manually added as a separate Inbound rule
ANSWER: C

How does Amazon Elastic Block Store (EBS) differ from EC2 Instance Store (ephemeral storage) when attached to an Amazon EC2 instance?
A. EBS is network-attached persistent block storage that survives stopping and restarting the EC2 instance independently, whereas Instance Store is physically attached to the underlying host server and loses all data if the instance is stopped, hibernated, or terminated
B. Instance Store supports point-in-time S3 snapshots automatically, whereas EBS does not
C. EBS data is wiped whenever the EC2 instance is stopped, whereas Instance Store data persists forever
D. EBS can only store static images accessed via HTTP URLs, whereas Instance Store is a serverless SQL database
ANSWER: A

An EC2 instance running a Python script needs to upload backup files to an Amazon S3 bucket (s3://cdac-lab-backups). According to AWS security best practices, how should you grant the EC2 instance permission to write to that S3 bucket?

A. Create an IAM root access key ID and secret access key and hardcode them inside /etc/profile on the EC2 instance
B. Turn off 'Block all public access' on the S3 bucket and grant anonymous write permissions to *
C. Copy the S3 bucket's SSH .pem private key onto the EC2 instance
D. Create an IAM Role with an attached least-privilege IAM policy allowing s3:PutObject on the target bucket ARN, and attach that IAM Role (via an Instance Profile) to the EC2 instance so the AWS SDK automatically retrieves temporary rotating credentials from the Instance Metadata Service (IMDSv2)
ANSWER: D

An organization stores 50 TB of compliance audit records in Amazon S3 that are almost never accessed, but if a regulatory audit occurs, they can wait 3 to 5 hours to retrieve the archives at the lowest possible storage cost. Which S3 Storage Class is specifically designed for this use case?
A. S3 One Zone-Infrequent Access
B. S3 Standard
C. S3 Glacier Flexible Retrieval (or S3 Glacier Deep Archive for 12-hour retrieval)
D. S3 Intelligent-Tiering (Frequent Access tier)
ANSWER: C

When an AWS Lambda function is triggered synchronously by Amazon API Gateway or asynchronously by an Amazon S3 ObjectCreated event, what happens during a 'Cold Start', and which IAM resource defines the permissions that the Lambda function itself has to access other AWS services (such as writing logs to Amazon CloudWatch Logs)?
A. Lambda functions never experience Cold Starts and do not require IAM permissions to write CloudWatch logs
B. During a Cold Start, AWS reboots the entire Availability Zone; permissions are defined by the EC2 Key Pair
C. During a Cold Start, Lambda compiles C++ kernel drivers; permissions are defined by the S3 Bucket ACL
D. During a Cold Start, AWS provisions a new execution environment (microVM), downloads the function code/layers, and initializes the runtime and global handler code before invoking the handler; the function's permissions to write to CloudWatch Logs (logs:CreateLogGroup, logs:CreateLogStream, logs:PutLogEvents) are defined by its IAM Execution Role
ANSWER: D

What is an Amazon Virtual Private Cloud (AWS VPC)?
A. A logically isolated virtual network dedicated to your AWS account where you can launch AWS resources in a custom-defined IP address range (CIDR block), subnets, route tables, and network gateways
B. A managed Active Directory domain controller
C. A physical hardware firewall shipped to a customer's office
D. A serverless content delivery network for caching HTML pages
ANSWER: A

In the Session 20 & 21 Lab Assignment, you create a VPC named ditiss-lab with the network CIDR 172.20.0.0/16, a public subnet 172.20.5.0/24, and a private subnet 172.20.10.0/24. How many total IPv4 addresses are defined in a /24 subnet block, and how many of those IPv4 addresses does AWS reserve in every VPC subnet (making them unavailable for EC2 assignment)?
A. 128 total IPv4 addresses; AWS reserves 0 addresses
B. 256 total IPv4 addresses; AWS reserves 5 addresses in every subnet (the first 4 and the last 1, leaving 251 usable host IP addresses)
C. 256 total IPv4 addresses; AWS reserves 2 addresses (leaving 254 usable)
D. 65,536 total IPv4 addresses; AWS reserves 10 addresses
ANSWER: B

In the ditiss-lab VPC (172.20.0.0/16), what specifically makes the subnet 172.20.5.0/24 a 'Public Subnet' rather than a 'Private Subnet'?
A. Placing the subnet in a different AWS Region than the VPC
B. Associating the subnet with a Route Table that has a route for internet-bound traffic (0.0.0.0/0) pointing to an attached Internet Gateway (igw-xxxxxxxx)
C. Disabling all Security Groups on instances inside 172.20.5.0/24
D. Naming the subnet with the word public in the AWS Console Tag field
ANSWER: B

In the ditiss-lab Lab Assignment, specifically in the private subnet 172.20.10.0/24, which exact five IPv4 addresses are reserved by AWS and cannot be assigned to an EC2 instance?
A. 172.20.10.10 through 172.20.10.14
B. 172.20.0.0 through 172.20.0.4

C. 172.20.10.0 (Network address), 172.20.10.1 (VPC router), 172.20.10.2 (AWS DNS server), 172.20.10.3 (Reserved for future use), and 172.20.10.255 (Network broadcast address)
D. 172.20.10.0, 172.20.10.1, 172.20.10.253, 172.20.10.254, and 172.20.10.255
ANSWER: C

In the ditiss-lab Lab Assignment, you launch one EC2 instance in the private subnet (172.20.10.0/24) and one EC2 instance in the public subnet (172.20.5.0/24). Since the private instance has no Public IPv4 address, how can the private instance connect outbound to the internet to run yum install httpd or apt install apache2 without accepting unsolicited inbound traffic from the internet?
A. By assigning the broadcast IP 172.20.10.255 to the private EC2 instance
B. By disabling the VPC router route 172.20.0.0/16 -> local
C. By attaching a second Internet Gateway directly to the private subnet
D. By provisioning a NAT Gateway in the public subnet (172.20.5.0/24) with an Elastic IP, and adding a default route (0.0.0.0/0 -> nat-gateway-id) in the private subnet's (172.20.10.0/24) Route Table
ANSWER: D

In the ditiss-lab Lab Assignment, once httpd is installed and started on the private subnet instance (e.g., private IP 172.20.10.25), you log into the public subnet instance (172.20.5.0/24) and run curl https://www.google.com/search?q=http://172.20.10.25. Which VPC routing rule allows the public subnet instance and private subnet instance to communicate directly with each other, and what Security Group rule is needed on the private instance?
A. An Internet Gateway route 0.0.0.0/0 routes the traffic; the private instance needs TCP port 22 open to 0.0.0.0/0
B. A BGP VPN tunnel is required between two subnets in the same VPC; the private instance needs UDP port 53 open
C. Traffic between 172.20.5.0/24 and 172.20.10.0/24 is blocked by AWS hardware unless VPC Peering is configured
D. The default VPC Route Table entry 172.20.0.0/16 -> local routes traffic internally between all subnets in ditiss-lab; the private instance's Security Group must allow Inbound HTTP (TCP port 80) from 172.20.5.0/24 (or from the public instance's Security Group ID)
ANSWER: D

How does an AWS Network Access Control List (NACL) differ from an AWS Security Group in a VPC?
A. NACLs only filter IPv6 traffic, whereas Security Groups only filter IPv4 traffic
B. NACLs operate at the subnet boundary, are stateless (requiring explicit inbound AND outbound ephemeral port rules), and support both Allow and Deny rules evaluated in rule-number order; Security Groups operate at the instance ENI level, are stateful, and support Allow rules only
C. NACLs cannot be modified after a VPC is created
D. NACLs apply at the EC2 instance NIC level and are stateful, whereas Security Groups apply at the subnet level and are stateless
ANSWER: B

In the ditiss-lab Lab Assignment, a student wants to SSH from their laptop into the private EC2 instance (172.20.10.25) using the public EC2 instance (172.20.5.15) as a Bastion Host (Jump Host). Instead of copying the .pem private key file onto the public Bastion Host's disk (which is a security risk if the bastion is compromised), what SSH command or flag from the laptop securely forwards authentication or jumps through the bastion host?
A. ssh -p 80 ec2-user@172.20.10.25
B. ssh -J ec2-user@<public-ip> ec2-user@172.20.10.25 -i lab-key.pem (ProxyJump) or SSH Agent Forwarding (ssh -A)
C. telnet <public-ip> 22 && curl https://www.google.com/search?q=http://172.20.10.25
D. scp lab-key.pem /var/www/html/ && ssh 172.20.10.25
ANSWER: B

In the ditiss-lab Lab Assignment, the httpd service is confirmed running (systemctl status httpd is active) on the private instance (172.20.10.25), and the private instance's AWS Security Group allows Inbound TCP port 80 from 172.20.5.0/24. However, running curl https://www.google.com/search?q=http://172.20.10.25 from the public instance (172.20.5.15) hangs and times out because a custom NACL was attached to the private subnet that only has Inbound Rule 100 (Allow TCP 80 from 172.20.5.0/24) and an Outbound Deny All rule. Why does curl time out, and what NACL rule is missing?
A. Because NACLs are stateless: although the HTTP SYN request enters the private subnet on TCP port 80, the outbound HTTP response from 172.20.10.25 back to the client's high ephemeral port (1024-65535) on 172.20.5.0/24 is blocked by the Outbound NACL
B. Because httpd listens on port 443 by default when started on a private subnet
C. Because curl uses UDP port 80 by default
D. Because 172.20.0.0/16 is a public routable Class A address block

ANSWER: A

If an EC2 instance inside the private subnet (172.20.10.0/24) needs to access an Amazon S3 bucket or DynamoDB table privately over the AWS internal network without routing traffic through a NAT Gateway or Internet Gateway, which VPC feature should be configured?
A. An AWS CloudFront Distribution attached to 172.20.10.0/24
B. A VPC Gateway Endpoint for S3/DynamoDB (which adds a prefix-list route directly to the private subnet's Route Table at no hourly gateway cost)
C. An AWS Direct Connect physical fiber circuit
D. An Elastic Network Adapter with promiscuous mode enabled
ANSWER: B

Which command initializes the current Docker host as the first Manager node of a new Docker Swarm cluster?
A. docker node create --manager
B. docker cluster start
C. docker swarm init
D. docker service init
ANSWER: C

After initializing a Docker Swarm manager node, which command generates and displays the exact docker swarm join --token ... command needed to add a new Worker node to the cluster?
A. docker swarm connect
B. docker node add worker
C. docker service token --worker
D. docker swarm join-token worker
ANSWER: D

How does a Microservices architecture differ from a Monolithic application architecture?
A. Microservices require all components to share a single global database table and be written in the exact same programming language
B. Microservices cannot be deployed using container orchestrators
C. A Monolithic application deploys every feature as a separate container across different cloud providers, whereas Microservices compile everything into one .exe binary
D. A Microservices architecture decomposes an application into loosely coupled, independently deployable services organized around specific business capabilities (communicating via lightweight APIs/messaging), whereas a Monolith packages UI, business logic, and data access into a single tightly coupled deployable unit
ANSWER: D

In the Session 22 & 23 Lab Assignment: 'Create a new docker image using NGINX. Push it to docker hub. Create a service on docker swarm. Keep initial replicas as 1...', which command creates a Docker Swarm service named web-svc publishing port 8080 to container port 80 with 1 initial replica from myuser/nginx:v1?
A. docker run --swarm --name web-svc --replicas 1 -p 8080:80 myuser/nginx:v1
B. docker swarm deploy --name web-svc --count 1 myuser/nginx:v1
C. docker stack create web-svc --replicas 1 myuser/nginx:v1
D. docker service create --name web-svc --replicas 1 -p 8080:80 myuser/nginx:v1
ANSWER: D

In the Docker Swarm Lab Assignment, after creating the web-svc service with 1 replica, which commands scale the service up to 10 replicas and then scale it down to 2 replicas?
A. docker service scale web-svc=10 and docker service scale web-svc=2 (or docker service update --replicas 10 web-svc and --replicas 2)
B. docker service resize web-svc 10 and docker service resize web-svc 2
C. docker container scale web-svc 10 and docker container scale web-svc 2
D. docker swarm replicas web-svc 10 and docker swarm replicas web-svc 2
ANSWER: A

In the Docker Swarm Lab Assignment, after modifying index.html, building a new image myuser/nginx:v2, and pushing it to Docker Hub, which command performs a rolling update of the existing web-svc Swarm service to use the new myuser/nginx:v2 image?
A. docker pull myuser/nginx:v2 --service web-svc
B. docker service replace web-svc myuser/nginx:v2
C. docker swarm upgrade web-svc myuser/nginx:v2
D. docker service update --image myuser/nginx:v2 web-svc
ANSWER: D

What is the difference between a replicated service mode and a global service mode in Docker

Swarm?
A. A global service exposes ports to the public internet, whereas a replicated service is private
B. A replicated service cannot be updated without deleting the Swarm cluster
C. A replicated service runs only on Manager nodes, whereas a global service runs only on Worker nodes
D. A replicated service schedules a specified number of identical task replicas across the cluster (--replicas N), whereas a global service (--mode global) automatically runs exactly one task replica on every available node in the Swarm cluster (ideal for monitoring agents like Node Exporter)
ANSWER: D

When a Docker Swarm service publishes port 8080 (-p 8080:80) with --replicas 2 across a 5-node Swarm cluster, a user sends an HTTP request to port 8080 on one of the 3 Swarm nodes that is NOT currently running any container replica of that service. How does Docker Swarm handle this request?
A. The connection is refused (Connection reset by peer) because no replica is running on that specific node
B. Docker Swarm immediately spins up a 3rd replica on that node before answering the TCP SYN packet
C. The request is redirected via HTTP 302 to Docker Hub
D. Docker Swarm's built-in Ingress Routing Mesh accepts the request on port 8080 on ANY node in the cluster and transparently load-balances and routes the packet over the internal ingress overlay network (via IPVS) to an active container replica on another node
ANSWER: D

Docker Swarm Manager nodes use the Raft Consensus Algorithm to maintain cluster state. Why is a 3-manager or 5-manager configuration strongly recommended over a 2-manager configuration, and how many manager node failures can a 3-manager cluster versus a 5-manager cluster tolerate while maintaining quorum?
A. Both 3-manager and 5-manager clusters can tolerate all but 1 manager failing
B. A 2-manager cluster can tolerate 1 failure, whereas a 3-manager cluster can tolerate 2 failures
C. Raft requires a strict majority quorum of (N/2) + 1 managers to agree on cluster state; a 2-manager cluster has a quorum of 2 (tolerating 0 failures, making it less reliable than 1 manager), whereas a 3-manager cluster tolerates 1 manager failure (quorum 2) and a 5-manager cluster tolerates 2 manager failures (quorum 3)
D. Raft only supports even numbers of nodes unless an external ZooKeeper cluster is installed
ANSWER: C

When executing docker service update --image myuser/nginx:v2 web-svc on a Swarm service with 10 replicas, which flags control how many replicas are updated simultaneously, the delay between batch updates, and automatic rollback if the new image fails health checks?
A. --batch-size, --sleep, and --undo
B. --update-parallelism, --update-delay, and --update-failure-action rollback (or docker service rollback web-svc)
C. --threads, --timeout, and --revert
D. --max-surge, --max-unavailable, and --canary
ANSWER: B

In Kubernetes, what is the smallest and simplest deployable compute unit that encapsulates one or more tightly coupled containers sharing the same network namespace (IP address and port space) and storage volumes?
A. Node
B. ReplicaSet
C. Pod
D. StatefulSet
ANSWER: C

Which command-line tool is used by administrators and developers to interact with the Kubernetes API Server to deploy applications, inspect cluster resources, and view logs?
A. kubectl
B. kubelet
C. kube-proxy
D. kubeadm
ANSWER: A

In a Kubernetes cluster architecture, which four core components run on the Control Plane (Master Node)?
A. kube-apiserver, etcd, kube-scheduler, and kube-controller-manager
B. kubelet, kube-proxy, containerd, and nginx

C. dockerd, swarmd, prometheus, and grafana
D. chef-server, puppetserver, ansible, and terraform
ANSWER: A

In the Session 22 & 23 Lab Assignment: 'Configure a Linux as a Master node using Kubernetes and make 2 to 3 worker node clusters', which two primary node components must run on every Worker node (in addition to a Container Runtime like containerd) to register the node, manage Pod lifecycles, and maintain network routing rules?
A. helm and minikube
B. etcd and kube-scheduler
C. kube-apiserver and coredns
D. kubelet and kube-proxy
ANSWER: D

When bootstrapping a multi-node Kubernetes cluster on Linux machines using kubeadm (1 Master node and 2 to 3 Worker nodes), which Linux system configuration prerequisite must be disabled on all nodes before running kubeadm init so the kubelet starts properly?
A. IPv4 packet forwarding (net.ipv4.ip_forward)
B. The OpenSSH daemon (sshd)
C. Linux Swap memory (swapoff -a and commenting out swap in /etc/fstab)
D. The loopback network interface (lo)
ANSWER: C

In the Lab Assignment where you repeat the NGINX deployment lifecycle on Kubernetes (start with 1 replica, increase replicas to 10, reduce replicas to 2, and update the deployment to a new modified NGINX image myuser/nginx:v2), which sequence of kubectl commands accomplishes these steps for a deployment named nginx-deploy?
A. kubectl create deployment nginx-deploy --image=myuser/nginx:v1 --replicas=1 -> kubectl scale deployment nginx-deploy --replicas=10 -> kubectl scale deployment nginx-deploy --replicas=2 -> kubectl set image deployment/nginx-deploy nginx=myuser/nginx:v2
B. kubectl service create nginx-deploy -> kubectl service scale 10 -> kubectl service update
C. kubeadm deploy nginx-deploy -> kubeadm scale 10 -> kubeadm scale 2
D. kubectl run nginx-deploy --image=myuser/nginx:v1 -> kubectl resize 10 -> kubectl resize 2 -> kubectl upgrade
ANSWER: A

In Kubernetes, Pods are ephemeral and their IP addresses change whenever they are recreated. Which Kubernetes resource provides a stable virtual IP address and DNS name to expose the nginx-deploy Pods, and which Service type exposes the service on a static port (in the default range 30000-32767) across every node's IP in the cluster?
A. A PersistentVolumeClaim of type HostPath
B. A ConfigMap of type ExternalName
C. A Kubernetes Service of type NodePort (created via kubectl expose deployment nginx-deploy --type=NodePort --port=80)
D. A DaemonSet of type ClusterIP
ANSWER: C

After running kubeadm init --pod-network-cidr=10.244.0.0/16 on the Linux Master node and copying /etc/kubernetes/admin.conf to ~/.kube/config, running kubectl get nodes shows the Master node status as NotReady, and coredns Pods remain in Pending state. What step is required to transition the nodes to Ready and allow Pod-to-Pod networking across the Master and 2â€“3 Worker nodes?
A. Reboot the Master node into single-user recovery mode
B. Install a Container Network Interface (CNI) plugin add-on (such as Flannel, Calico, or Weave Net) using kubectl apply -f <cni-manifest.yaml>
C. Run kubeadm reset on the Master node
D. Format the worker nodes with ZFS
ANSWER: B

When you execute kubectl set image deployment/nginx-deploy nginx=myuser/nginx:v2 to update the NGINX deployment to the modified index.html image, how does the Kubernetes Deployment controller perform a zero-downtime RollingUpdate under the hood, and which command checks the status or rolls back the deployment if v2 is broken?
A. It deletes all existing Pods first, waits 60 seconds, and creates new Pods directly without using a ReplicaSet
B. It edits the running container's /usr/share/nginx/html/index.html file in-place via docker cp
C. The Deployment creates a new ReplicaSet for myuser/nginx:v2 and gradually scales it up while scaling down the old ReplicaSet (myuser/nginx:v1) according to maxSurge and maxUnavailable; monitored with kubectl rollout status deployment/nginx-deploy and rolled back with kubectl rollout undo deployment/nginx-deploy

D. It requires etcd to be stopped and restarted on all worker nodes
ANSWER: C

In Kubernetes, what is the role of etcd on the Control Plane, and why do production Kubernetes clusters configure both livenessProbe and readinessProbe on microservice Pods?
A. etcd is the consistent, distributed key-value store that holds all cluster state and configuration data; a livenessProbe detects deadlocked containers so kubelet can restart them, whereas a readinessProbe determines when a Pod is ready to accept traffic so the Service endpoints remove unready Pods from load balancing without killing them
B. etcd schedules Pods onto nodes; livenessProbe scales deployments to 10 replicas
C. etcd runs on every worker node to proxy TCP packets; both probes perform identical container restarts
D. etcd is a container image registry; livenessProbe checks disk space and readinessProbe checks CPU temperature
ANSWER: A

How does Ansible's architecture differ from traditional configuration management tools like Chef and Puppet?
A. Ansible is agentless: it does not require installing a persistent background agent on managed Linux nodes, communicating instead over standard OpenSSH (and Python) from the Control Node using a push model
B. Ansible requires a heavy Java daemon running on every managed target node listening on TCP port 8140
C. Ansible Configuration files are written in compiled C++ binaries
D. Ansible can only manage Windows Vista workstations and does not support Linux
ANSWER: A

In the Session 24 & 25 Lab Assignment for setting up the Ansible environment, which two commands are run on the Ansible Control Node to (1) generate an SSH key pair and (2) copy the public key to a target node (user@192.168.1.20) for passwordless SSH authentication?
A. ansible-keygen and ansible-copy user@192.168.1.20
B. openssl genrsa and scp /etc/ssh/sshd_config user@192.168.1.20
C. ssh-keygen and ssh-copy-id user@192.168.1.20
D. ssh-create and ssh-push user@192.168.1.20
ANSWER: C

In the Lab Assignment command ansible -m ping all, what do the -m ping option and the all argument specify, and what does the ping module actually verify?
A. -m ping sends raw ICMP echo requests; all pings every IP on the internet
B. -m ping restarts the network card on the local machine; all clears the ARP cache
C. -m ping invokes the built-in Ansible ping module across all hosts listed in the inventory; it verifies end-to-end SSH connectivity, authentication, and the presence of a usable Python interpreter on the target nodes (returning "ping": "pong")
D. -m ping installs the iputils-ping package via apt on the Control Node
ANSWER: C

What is the default file path on Linux for the system-wide static Ansible inventory file, and what flag is passed to ansible or ansible-playbook to specify a custom static inventory file in the current project directory?
A. /etc/ansible/hosts; specified with -i <inventory-file>
B. /usr/local/etc/inventory.cfg; specified with -h
C. ~/.ansible_hosts; specified with -c
D. /var/lib/ansible/nodes.json; specified with -f
ANSWER: A

In a static INI-formatted Ansible inventory file, how do you define a host group named webservers containing two nodes, and how do you define a parent group named production that includes the webservers and dbservers groups?
A. Use [webservers] for the host group and [production:children] to group webservers and dbservers
B. Use {webservers} and {production:parent}
C. Use <webservers> and <production extends="webservers">
D. Use group: webservers and include: dbservers
ANSWER: A

What does 'Idempotency' mean in the context of Ansible modules?
A. Running an Ansible task multiple times appends duplicate configuration lines to target files every time
B. An operation can be applied once or multiple times and will always result in the exact same desired system state without making unintended changes if the target is already in that desired

state (reporting ok instead of changed)
C. Ansible executes tasks in a random order on every run to test chaos engineering
D. Modules automatically uninstall packages 24 hours after installation
ANSWER: B

When running an Ansible ad-hoc command (ansible webservers -m apt -a "name=nginx state=present") as a non-root SSH user (ubuntu) who has sudo privileges on the target nodes, which flag must be added to escalate privileges to root?
A. --su-admin
B. --privileged
C. --root
D. -b (or --become, optionally with --ask-become-pass / -K if passwordless sudo is not configured)
ANSWER: D

What is the precedence order in which Ansible searches for its configuration file (ansible.cfg) when executing a command?
A. 1. /etc/ansible/ansible.cfg -> 2. ~/.ansible.cfg -> 3. ./ansible.cfg -> 4. ANSIBLE_CONFIG
B. Ansible only reads /etc/ansible/ansible.cfg and ignores local directories
C. 1. ~/.bashrc -> 2. /etc/profile -> 3. /var/log/ansible.cfg
D. 1. ANSIBLE_CONFIG environment variable -> 2. ./ansible.cfg (in the current working directory) -> 3. ~/.ansible.cfg (in the user's home directory) -> 4. /etc/ansible/ansible.cfg
ANSWER: D

In an Ansible project directory, instead of cluttering the static inventory file with dozens of host-specific and group-specific variables, which two standard subdirectories are automatically searched by Ansible to load YAML variable files matching inventory group names and hostnames?
A. group_vars/ and host_vars/
B. vars_group/ and vars_host/
C. env_vars/ and node_vars/
D. defaults/ and meta/
ANSWER: A

On a brand-new server deployment, running ansible -m ping all interactively prompts Are you sure you want to continue connecting (yes/no/[fingerprint])? and stalls across 20 target nodes, OR fails on a minimal Ubuntu image that only has /usr/bin/python3 missing legacy /usr/bin/python. Which configuration settings resolve these two issues?
A. Run ansible -m raw reboot all before every ping command
B. Set transport = ftp and install Java 17 on the Control Node
C. Disable SSH keys and use Telnet on port 23
D. Configure host_key_checking = False in ansible.cfg (or pre-populate ~/.ssh/known_hosts with ssh-keyscan) and set ansible_python_interpreter=/usr/bin/python3 (or auto) in inventory/config
ANSWER: D

Which data serialization format is used to write Ansible Playbooks, and which command-line tool executes an Ansible Playbook?
A. JSON format; executed with ansible-run
B. Ruby DSL; executed with ansible-apply
C. YAML format (.yml / .yaml); executed with ansible-playbook
D. XML format; executed with ansible
ANSWER: C

Which command-line tool bundled with Ansible generates the standardized directory skeleton (tasks/, handlers/, templates/, files/, vars/, defaults/, meta/) for a new reusable Ansible Role named mysql_db?
A. ansible-playbook --create-role mysql_db
B. ansible-vault create mysql_db
C. ansible-doc init mysql_db
D. ansible-galaxy init mysql_db (or ansible-galaxy role init mysql_db)
ANSWER: D

In the Lab Assignment to write an Ansible Playbook that installs and starts a web server (Apache or Nginx) and ensures it starts at boot on a target Linux node, which Ansible module manages system services (state: started, enabled: yes) across modern systemd-based Linux distributions?
A. ansible.builtin.service (or ansible.builtin.systemd)
B. ansible.builtin.copy
C. ansible.builtin.debug
D. ansible.builtin.lineinfile
ANSWER: A

When writing an Ansible Playbook in YAML, which formatting rule is strictly required for valid YAML syntax, and which command checks a playbook for syntax errors without executing it?
A. Playbooks must be wrapped in { } curly braces; checked with ansible -m check
B. Indentation must use Tab (\t) characters only; checked with ansible-playbook --test
C. Every line must end with a semicolon (;); checked with ansible-lint --compile
D. Indentation must use spaces (never Tab characters) with consistent alignment for sibling keys; checked with ansible-playbook --syntax-check playbook.yml
ANSWER: D

Before applying a new Ansible Playbook to production web servers, an administrator wants to perform a dry run that predicts and reports what changes would be made on the target nodes without actually modifying anything. Which flag should be passed to ansible-playbook?
A. --noop
B. --plan
C. --dry
D. --check (or -C, optionally paired with --diff / -D)
ANSWER: D

In an Ansible Playbook or Role, how do handlers work in combination with the notify keyword (for example, when modifying /etc/nginx/nginx.conf or /etc/mysql/my.cnf)?
A. A handler is a special task triggered via notify: ONLY when the notifying task actually results in a changed state on the target node, and handlers execute once at the end of the play even if notified by multiple tasks
B. Handlers run exclusively on the Ansible Control Node to send emails
C. Handlers automatically delete the modified configuration file if a syntax error occurs
D. Handlers run before every task in the playbook regardless of whether any file changed
ANSWER: A

In the Lab Assignment to develop a role to install and configure a database server (MySQL or PostgreSQL) and apply it via a playbook, which file inside the role (roles/dbserver/...) serves as the main entry point containing the list of tasks that Ansible automatically executes when the role is invoked?
A. roles/dbserver/playbook.yml
B. roles/dbserver/tasks/main.yml
C. roles/dbserver/vars/tasks.yml
D. roles/dbserver/tasks/init.yml
ANSWER: B

An administrator writes a single portable Ansible Playbook to install and start Apache across a mixed inventory containing both Ubuntu (Debian family, package/service name apache2) and Rocky Linux (RedHat family, package/service name httpd) target nodes. Which automatically gathered Ansible Fact variable and conditional keyword allow tasks to adapt to the target OS family?
A. Fact ohai_platform with the unless: keyword
B. Fact os.type with the switch: keyword
C. Fact facter_kernel with the if: keyword
D. Fact ansible_facts['os_family'] (or ansible_os_family) evaluated with the when: conditional keyword (e.g., when: ansible_os_family == "Debian")
ANSWER: D

Within an Ansible Role, variables can be defined in either defaults/main.yml or vars/main.yml. What is the difference in variable precedence between these two directories, and which templating engine does the ansible.builtin.template module use to render dynamic .j2 configuration files?
A. defaults/main.yml has the lowest precedence of all variable locations (making it easy to override in inventory or playbooks), whereas vars/main.yml has high role-level precedence; the template module uses the Jinja2 (.j2) templating engine
B. defaults/main.yml is only read on Windows nodes, whereas vars/main.yml is read on Linux nodes
C. defaults/main.yml has the highest precedence in Ansible; templates use Embedded Ruby (.erb)
D. Both directories have identical precedence; templates use Mustache syntax
ANSWER: A

When developing the Ansible Role to install and configure MySQL or PostgreSQL in the Lab Assignment, the playbook needs to store the database root password securely in version control without exposing it in plaintext. Which built-in Ansible tool encrypts variable files or strings using AES-256 encryption, and which flag is passed to ansible-playbook to decrypt them at runtime?
A. ansible-vault (e.g., ansible-vault create / ansible-vault encrypt); decrypted at runtime with --ask-vault-pass or --vault-password-file
B. ansible-secret; decrypted with --unlock
C. gpg-ansible; decrypted with --passphrase
D. ansible-crypt; decrypted with --key

ANSWER: A

What is Terraform, and what configuration language does it use by default to declare cloud and on-premises infrastructure resources?
A. An open-source declarative Infrastructure as Code (IaC) provisioning tool created by HashiCorp that uses HashiCorp Configuration Language (HCL, in .tf files) or JSON
B. An imperative procedural shell-scripting framework written in Python
C. An AWS-proprietary service that replaces CloudFormation and only runs inside AWS Lambda
D. A container runtime engine that replaces Docker and Kubernetes
ANSWER: A

In the standard Terraform workflow, what is the first command you must run inside a new directory containing .tf configuration files to initialize the working directory, download required provider plugins (such as hashicorp/aws), and configure the backend?
A. terraform plan
B. terraform setup
C. terraform init
D. terraform start
ANSWER: C

Which three core Terraform commands represent the standard lifecycle to (1) preview the execution plan of changes Terraform will make, (2) provision/modify the actual infrastructure resources in AWS, and (3) tear down all managed resources when finished?
A. terraform check, terraform run, and terraform delete
B. terraform preview, terraform deploy, and terraform remove
C. terraform plan, terraform apply, and terraform destroy
D. terraform diff, terraform push, and terraform clean
ANSWER: C

In DevOps automation, how does an Infrastructure Provisioning tool like Terraform complement a Configuration Management tool like Ansible?
A. Terraform and Ansible cannot be used together in the same project
B. Terraform is a declarative, state-aware infrastructure orchestrator designed to provision and manage the lifecycle of immutable cloud infrastructure (VPCs, subnets, security groups, load balancers, EC2 instances), whereas Ansible is primarily used to configure software, OS packages, and services inside running OS instances
C. Terraform can only manage on-premises switches, whereas Ansible can only create AWS accounts
D. Ansible requires a state file (ansible.tfstate) to track cloud API resources, whereas Terraform is completely stateless
ANSWER: B

In the Session 26 & 27 Lab Assignment for configuring AWS credentials so Terraform can interact with AWS services on a local machine, what is the standard secure way to provide credentials instead of hardcoding access_key and secret_key directly inside .tf files?
A. Pass the AWS password over unencrypted Telnet to https://www.google.com/search?q=sts.amazonaws.com
B. Run aws configure to store credentials in ~/.aws/credentials (or export AWS_ACCESS_KEY_ID and AWS_SECRET_ACCESS_KEY environment variables / assume an IAM Role), which the Terraform AWS provider automatically detects
C. Store credentials inside /etc/hosts
D. Write the secret key in the README.md file of the repository
ANSWER: B

In the Lab Assignment to create a basic Terraform configuration that provisions an EC2 instance in AWS, which HCL block type and resource type are used, and which two arguments are minimally required inside that resource block?
A. data "aws_ec2" "web" requiring vpc_id and key_name
B. module "aws_vm" "web" requiring cpu and ram
C. provider "ec2_server" "web" requiring region and zone
D. resource "aws_instance" "web" requiring ami and instance_type
ANSWER: D

Which built-in Terraform CLI commands are used to (1) automatically reformat .tf configuration files to canonical HCL style and (2) verify whether a configuration is syntactically valid and internally consistent without accessing remote cloud APIs?
A. terraform lint and terraform test-syntax
B. terraform beauty and terraform verify
C. terraform style and terraform check
D. terraform fmt and terraform validate

ANSWER: D

When terraform init is executed in a project directory, it downloads the specified provider binary into .terraform/providers/ and creates a file named .terraform.lock.hcl. What is the purpose of .terraform.lock.hcl, and should it be committed to version control (Git)?
A. It locks the AWS console so no human can log in; it must be deleted after every run
B. It contains the plaintext AWS secret access keys and state outputs; it must NEVER be committed to Git
C. It is a temporary swap file created only when terraform apply crashes
D. It is the dependency lock file that records exact provider versions and cryptographic hashes to ensure every team member and CI/CD runner uses the exact same provider binaries; it SHOULD be committed to Git
ANSWER: D

An aws_instance resource is currently running in AWS with instance_type = "t2.micro" and ami = "ami-01111111". If you change instance_type to "t3.small" in main.tf, versus changing ami to "ami-02222222", how does terraform plan handle each modification?
A. Changing instance_type deletes the entire VPC, whereas changing ami updates the instance in-place without rebooting
B. Changing instance_type performs an in-place update (~ stopping, resizing, and starting the existing instance ID), whereas changing ami forces a destructive replacement (-/+ destroying the existing EC2 instance and provisioning a brand-new instance ID)
C. Both changes require manually running terraform destroy first
D. Both changes are ignored by Terraform because EC2 instances are immutable after creation
ANSWER: B

In Terraform HCL, what is the difference between a resource block (e.g., resource "aws_ami" "custom") and a data block (e.g., data "aws_ami" "ubuntu")?
A. There is no difference; data is just an alias for resource in Terraform v1.x
B. A resource block is read-only, whereas a data block creates and destroys cloud infrastructure
C. A data block stores SQL rows inside Terraform state, whereas a resource block stores S3 files
D. A resource block declares infrastructure objects that Terraform creates, modifies, and destroys, whereas a data source block is read-only and queries the provider API to fetch information about existing infrastructure or external resources managed outside the current Terraform configuration
ANSWER: D

In a well-organized Terraform project directory, what is the conventional split of responsibilities across main.tf, variables.tf, and outputs.tf?
A. variables.tf stores the state file, main.tf stores AWS credentials, and outputs.tf stores debug logs
B. outputs.tf is generated automatically by terraform init and cannot be edited by hand
C. Terraform only reads main.tf and ignores all other .tf files unless imported with #include
D. Terraform automatically loads all .tf files in the working directory: variables.tf declares input parameters (variable blocks), main.tf defines providers and core resource/module blocks, and outputs.tf declares exported return values (output blocks)
ANSWER: D

What is the default filename in which Terraform stores the local state mapping between your .tf configuration resources and real-world objects provisioned in the cloud?
A. terraform.tfstate
B. main.state.yml
C. .terraform.db
D. terraform.lock.json
ANSWER: A

If you declare an input variable variable "instance_type" { default = "t2.micro" } in variables.tf, what is the correct HCL syntax to reference this variable's value inside a resource block in main.tf?
A. var.instance_type
B. $instance_type
C. {{ instance_type }}
D. variable.instance_type.value
ANSWER: A

In the Session 26 & 27 Lab Assignment: 'Write a Terraform configuration file to provision a VPC, subnet, and an EC2 instance within the subnet', given resource "aws_vpc" "lab_vpc" and resource "aws_subnet" "lab_subnet", how do you link the subnet to the VPC and how does Terraform determine the creation order?

A. Run terraform apply three separate times while commenting out blocks manually
B. Use notify = aws_vpc.lab_vpc inside the EC2 block
C. Set vpc_id = aws_vpc.lab_vpc.id inside aws_subnet.lab_subnet and subnet_id = aws_subnet.lab_subnet.id inside aws_instance; these attribute references create implicit edges in Terraform's dependency graph so Terraform automatically provisions the VPC first, then the Subnet, and then the EC2 instance
D. Hardcode vpc_id = "vpc-12345" and use numbered filenames (01-vpc.tf, 02-subnet.tf)
ANSWER: C

When passing values to declared Terraform input variables without editing variables.tf, which file does Terraform automatically load if present in the working directory, and what happens if a variable has higher precedence via -var on the CLI?
A. vars.auto.json only; CLI -var flags have the lowest precedence
B. config.env; default attributes inside variables.tf override everything else
C. terraform.tfvars (or *.auto.tfvars); CLI -var and -var-file flags take highest precedence and override values set in terraform.tfvars or variable default attributes
D. inputs.yml; Terraform does not support command-line variable overrides
ANSWER: C

In the Lab Assignment: 'Create a Terraform module for provisioning an EC2 instance with a security group' placed in ./modules/ec2_sg, how does the root configuration call this child module, and how does the root configuration access the instance's public IP exported by the child module?
A. Run terraform module build ./modules/ec2_sg
B. Use a module "app_server" { source = "./modules/ec2_sg" ... } block (followed by terraform init), declare an output "public_ip" inside the child module, and reference it in the root module as module.app_server.public_ip
C. Use import "./modules/ec2_sg"; access via aws_instance.this.public_ip directly from the root
D. Copy and paste the .tf files into /etc/terraform/modules
ANSWER: B

In the Lab Assignment: 'Use the module in a Terraform configuration to provision multiple EC2 instances in different subnets', which two Terraform meta-arguments can be added directly inside a module (or resource) block to dynamically instantiate multiple instances across a list or map of subnet IDs without duplicating the module block?
A. repeat and while
B. count and for_each
C. replicas and scale
D. loop and with_items
ANSWER: B

In the Lab Assignment: 'Manage the Terraform state using a remote backend (e.g., AWS S3) to enable collaboration and state sharing', why should terraform.tfstate NEVER be committed to Git, and how does the AWS S3 remote backend prevent two engineers from running terraform apply simultaneously and corrupting the state file?
A. S3 automatically converts .tfstate into a CloudFormation stack; locking is handled by AWS Route 53
B. Engineers must email terraform.tfstate to each other before running terraform plan
C. terraform.tfstate contains sensitive values in plaintext and creates merge conflicts in Git; configuring a backend "s3" block inside terraform { ... } stores state centrally in S3 (with server-side encryption) and uses state locking (via an Amazon DynamoDB table dynamodb_table or S3 native lockfile use_lockfile = true in newer Terraform versions) to block concurrent state writes
D. terraform.tfstate is too large for Git; S3 prevents concurrent runs by disabling the internet gateway
ANSWER: C

When provisioning 3 EC2 instances across 3 different subnets (["subnet-a", "subnet-b", "subnet-c"]), why is using for_each = toset(var.subnet_ids) generally safer than using count = length(var.subnet_ids) if "subnet-a" (index 0) is later removed from the list?
A. With count, instances are indexed by numeric position ([0], [1], [2]), so removing index 0 shifts "subnet-b" to index 0 and "subnet-c" to index 1, forcing Terraform to recreate or modify all subsequent instances; with for_each, instances are keyed by their stable string identifier (["subnet-a"]), so only the removed "subnet-a" instance is destroyed while "subnet-b" and "subnet-c" remain untouched
B. count can only create a maximum of 2 resources
C. count does not work on module blocks in Terraform
D. for_each runs 10 times faster because it skips terraform plan
ANSWER: A

An administrator accidentally deletes an EC2 security group rule manually in the AWS Web Console

(causing configuration drift), or wants to inspect the resources currently tracked inside the remote S3 state without downloading the raw JSON file. Which commands list all resources tracked in the Terraform state and reconcile/detect real-world drift?

A. terraform ls to list state resources; terraform fix to repair drift

B. terraform inventory to list resources; terraform sync to overwrite AWS

C. terraform debug and terraform reset --hard

D. terraform state list (and terraform state show <address>) to inspect tracked state objects; terraform plan (or terraform apply -refresh-only) refreshes state against real-world AWS APIs to detect and reconcile drift

ANSWER: D